DIGITAL LEARNING CONTENT MANUAL

GRADE ELEVEN

ENGLISH
MATHS
BIOLOGY
CEMISTRY
PHYSICS
URDU

MINISTRY OF FEDERAL EDUCATION & PROFESSIONAL TRAINING

Table of Contents

1. Introduction

The Ministry of Federal Education and Professional Training is dedicated to enhancing the quality of distance learning and creating a comprehensive educational platform that serves a diverse group of learners across Pakistan. As part of this effort, the Ministry has procured digital content, including audio and video resources, that align with the Pakistan National Curriculum and Student Learning Outcomes (SLOs).

This guide is designed to help teachers navigate this digital content efficiently. Every video has been assigned a unique code, and this document organizes them in a way that clearly shows which video supports which SLO, for which grade and subject. Each table includes the video code, topic, SLO(s) covered, textbook unit, and duration.

Teachers can use this guide for lesson planning, in-class instruction, or revision sessions. It ensures that digital resources are used effectively, keeping lessons aligned with the curriculum and maximizing learning outcomes. The structure is consistent across all grades and subjects, making it easy to locate the right video at the right time.

This guide is intended as a practical companion to support teachers in delivering high-quality, curriculum-aligned instruction through digital content.

2. Available digital content from Grade KG- Grade 12

Digital Learning Content (Video)

Grade	Subject	No. of Packages	Total No. of Pacakges
KG	Math	21	133
	English	36	
	Urdu	25	
	General Knowledge	51	
Grade 1	Math	35	147
	English	33	
	Urdu	60	
	General Knowledge	19	
Grade 2	Math	45	166
	English	71	
	Urdu	16	
	General Knowledge	34	
Grade 3	English	92	176
	General Knowledge	29	
	Maths	39	
	Urdu	16	
Grade 4	English	92	190
	Math	36	
	Science	42	
	Urdu	20	
Grade 5	English	73	155
	Math	23	
	Science	34	
	Urdu	25	

Grade	Subject	No. of Packages	Total No. of Packages
Grade 6	Maths	31	120
	Science	27	
	English	37	
	Computer Science	25	
Grade 7	English	64	167
	Maths	51	
	Science	24	
	Computer Science	28	
Grade 8	English	60	162
	Maths	45	
	Science	25	
	Computer Science	32	

Total no. of Video Packages	2802

Grade	Subject	No. of Packages	Total No. of Pacakges
Grade 9	Biology	19	119
	Chemistry	21	
	English	20	
	Maths	42	
	Physics	17	
Grade 10	Biology	26	131
	Chemistry	12	
	English	21	
	Maths	55	
	Physics	17	
Grade 11	Biology	110	552
	Chemistry	94	
	English	55	
	Maths	88	
	Physics	91	
	Urdu	114	
Grade 12	Biology	103	544
	Chemistry	86	
	English	55	
	Maths	67	
	Pak. Studies	45	
	Physics	56	
	Urdu	132	
ECD – Grade 12	Gamified Educational Content	40	40

Sesame, Idara Taleem o Agahi (ITA), Environment & Entrepreneurship

Sesame Workshop	
Social Emotional Learning (SEL)	17
Maths	38
Science/ General Knowledge	21
Total	76 videos 55 (5 min videos) and 21 (27 minute videos)

ITA - Storytelling	
Grade 1	6
Grade 2	6
Grade 3	7
Total	19

Theme	No. of Packages
Environments for Living Lot 1 (ECD–2)	6
Environments for Living Lot 2 (Grades 3–5)	18
Environments for Living Lot 3 (Grades 6–8)	21
Entrepreneurship (Grades 6–8)	20

3. How to read video codes

Every video in this digital content collection has a unique code. The code helps teachers quickly identify:

- the grade
- the subject
- the textbook unit
- the package and part of the video

Understanding the code allows you to find the right video quickly without searching through multiple files.

Video Code Format

Example: **G01EnU01P01a**

- G01 = Grade 1
- En = English
- U01 = Unit 1
- P01a = Package 1, Part a

Code Reference Table:

Grade	Subject	Unit	Package
G01-G12, EY	M (Mathematics)	U01	P01
G01-G12, EY	Ur (Urdu)	U01	P01
G01-G12, EY	En (English)	U01	P01
G01-G12, EY	Sc (Science)	U01	P01
G01-G12, EY	Ph (Physics)	U01	P01
G01-G12, EY	Bi (Biology)	U01	P01
G01-G12, EY	Ch (Chemistry)	U01	P01
G12	PS (Pakistan Studies)	U01	P01

Notes for Teachers:

- Use this table to decode video codes quickly.
- “EY” refers to Early Years/Kindergarten.
- Package numbers indicate video sequence within a unit.
- Always cross-check the mapping tables in this guide for SLO coverage before showing a video in class.

4. Grade 11 Digital Content Alignment with SLOs

a. English

Unit	SLO	Package Code	Package Name
Competency 1: Reading and Thinking skills	➢ Apply strategies to comprehend questions by marking key words, verbs and tenses in a variety of question types: I. Literal/ textual/ factual ii. Interpretive iii. Inferential iv. Evaluative v. Personal response vi. Open ended ➢ Respond orally and in writing.	G11EnU01P01	Comprehension
	➢ Use dictionaries to • locate guide words. • locate entry word. • choose appropriate word definition. • identify pronunciation with the pronunciation key. • identify syllable division, and stress pattern. • identify parts of speech. • identify correct spellings. • identify phrases through key words. • recognize abbreviations used in a dictionary. • locate phrases and idioms. • comprehend notes on usage. • identify word etymology. ➢ Locate appropriate synonyms and antonyms in a thesaurus. ➢ Utilize appropriate informational sources including encyclopedias and internet sources. ➢ Use library skills to • alphabetize book titles, words and names. • locate fiction and nonfiction books / books by subject. • understand card catalogue. • locate and using card	G11EnU01P02	Correction of Sentences Part 1
		G11EnU01P03	Correction of Sentences Part 2
		G11EnU01P04	Correction of Sentences Part 3
		G11EnU01P05	Correction of Sentences Part 4
		G11EnU01P06	Correction of Sentences Part 5
		G11EnU01PO7	Correction of Sentences Part 6
		G11EnU01PO8	Correction of Sentences Part 7
		G11EnU01P09	Correction of Sentences Part 8
		G11EnU01P10	Correct Spellings

	catalogue. • identify three kinds of catalogue cards i.e. author card, title card, subject card. • use Dewey decimal system. • use numbers on books and catalogue cards. • use case and shelf labels in the library. • use the reference section in the library. • use computer catalogue. ➢ Utilize effective study strategies e.g. note taking / note making, writing a summary, creating a mind map to organize ideas. ➢ Use textual aids such as table of contents, footnote, glossary, preface, sub headings etc. to • comprehend texts. • identify and select relevant information in a book.		
Competency 2: Writing Skills	➢ Develop focus for his or her writing. ➢ Select and use a variety of pre-writing strategies such as brainstorming, mind mapping, outlining etc. ➢ Plan draft and revise writing to ensure that it • is focused, purposeful, includes a sense of audience, and shows insight into the writing situation. • has an appropriate writing style (expository, narrative, etc.) for a given purpose. • has an organizational pattern that reflects a clear overall progression of ideas through proper use of signal and reference words. • uses writing strategies as are appropriate to the purpose of writing. • has varied sentence structure and length. • has a good command of	G11EnU02P11	Subject Verb and Noun Agreement
		G11EnU02P12	Noun Pronoun Agreement and Modifiers
		G11EnU02P13	Parallelism and Wordiness
		G11EnU02P14	Essays Part 1
		G11EnU02P15	Essays Part 2
		G11EnU02P16	Essays Part 3
		G11EnU02P17	Essays Part 4
		G11EnU02P18	Essays Part 5

	language with precision of expression. ➢ Proof read and edit their own, peers', and given texts for errors of usage and style: • Faulty sentence structure. • Unclear pronoun reference. • Incomplete comparison. • Misplaced modifiers. • Dangling modifiers. • Subject / verb agreement. • Inconsistencies in verb / tense. • Faulty parallelism. • Confusion of adjectives and adverbs. • Wordy phrases. • Redundancy. • Vague language. • Inappropriate diction. • Clichés. • Conventions of format. • Errors of punctuation and spelling		
Competency 4: Formal and Lexical Aspects of Language.	➢ Use the pronunciation key to pronounce words with developing accuracy. ➢ Recognize silent letters in words and pronounce them with developing accuracy. ➢ Recognize, pronounce and represent primary and secondary stress in words with the help of a dictionary.	G11EnU04P19	Pronunciation
	➢ Use appropriate vocabulary and correct spelling in their own writing: • Illustrate the use of dictionary for finding appropriate meaning and correct spellings. • Use a thesaurus to locate the synonyms closest to the meaning of the given word in the context. • Examine and interpret transitional devices that show comparison, contrast, reason, concession, condition, emphasis. • Deduce the meaning of	G11EnU04P20	Idioms and Phrasal Verbs
		G11EnU04P21	Idioms Part 1
		G11EnU04P22	Idioms Part 2
		G11EnU04P23	Idioms Part 3
		G11EnU04P24	Idioms Part 4
		G11EnU04P25	Idioms Part 5
		G11EnU04P26	Idioms Part 6

	unfamiliar words from the context using contextual clues. • Use the knowledge of roots, suffixes and affixes to determine the meaning of unfamiliar words. • Understand and use colloquial and idiomatic expressions given in the text / glossary. • Explore the use of synonyms with varying shades of meaning used for various purposes e.g. propaganda, irony, parody and satire. • Use various reference sources to refine vocabulary for interpersonal, academic and work place situations, including figurative, idiomatic and technical vocabulary. ➢ Use the knowledge of literal and figurative meaning, grammatical gender and syntax to translate passages from English to Urdu. • Understand that most phrases and idioms do not translate literally from one language to another.	G11EnU04P27	Idioms Part 7
		G11EnU04P28	Idioms Part 8 and Vocabulary
		G11EnU04P29	Vocabulary, Morphology and Greek Root Knowledge
	➢ Apply rules of capitalization wherever applicable. ➢ Illustrate use of all punctuation marks wherever applicable. ➢ Recognize and rectify faulty punctuation in given passages and own work. ➢ Recognize and use comma to mark a dependent word or word group that breaks the continuity of the sentence. ➢ Recognize and use colon between two independent groups not joined by a connecting word, when the first group points forward to the second. ➢ Recognize and use semicolon: • Before certain expressions when they introduce an	G11EnU04P30	Punctuation Part 1
		G11EnU04P31	Punctuation Part 2
		G11EnU04P32	Punctuation Part 3

	illustration that is a complete clause or an enumeration that consists of several items. • To separate serial phrases or clauses which have a common dependence on something that precedes or follows. ➢ Recognize and use quotation marks to enclose titles of published works and titles of their subdivisions. ➢ Recognize and use hyphen to indicate the division of a word at the end of a line. ➢ Recognize and use dash to mark a parenthesis or apposition to give strong emphasis, to mark off a contrasting or summarizing statement. ➢ Recognize and use parenthesis (Square Brackets) to enclose explanation, comment or criticism inserted by someone other than the person quoted. ➢ Recognize and use omission marks or ellipses to signify the omission or deletion of letters or words in sentences.		
	➢ Demonstrate use of collective, countable and uncountable, material and abstract nouns. ➢ Demonstrate use of nouns, noun phrases and clauses in apposition. ➢ Apply rules of change of number of nouns. ➢ Recognize and demonstrate use of words that have double plurals	G11EnU04P33	Nouns and It's Types Part 1
		G11EnU04P34	Nouns and It's Types Part 2
		G11EnU04P35	Nouns and It's Types Part 3
		G11EnU04P36	Phrases and Clauses
	➢ Illustrate use of pronouns. ➢ Identify, and demonstrate use of relative pronouns. ➢ Recognize rules for using indefinite pronouns. ➢ Illustrate use of pronoun-antecedent agreement. ➢ Illustrate use of cataphoric and anaphoric references.	G11EnU04P37	Pronouns and It's Types Part 1
		G11EnU04P38	Pronouns and It's Types Part 2
		G11EnU04P39	Pronouns and It's Types Part 3

	➢ Apply rules for the use of a, an and the, wherever applicable in speech and writing	G11EnU04P40	Articles Part 1
		G11EnU04P41	Articles Part 2 and Verbs
	Illustrate the use and all functions of modal verbs. ➢ Illustrate use of regular and irregular verbs. ➢ Illustrate use of transitive and intransitive verbs. ➢ Make and use present and past participles. ➢ Identify, recognize the function and use of perfect participles. ➢ Illustrate the use of infinitives and infinitive phrases. ➢ Illustrate the use of gerunds and gerund phrases	G11EnU04P42	Types of Verbs
	➢ Illustrate use of tenses. ➢ Identify, change the form of, and use Future Continuous Tense. ➢ Identify form and use Future Continuous, Future Perfect and Future Perfect Continuous Tenses.	G11EnU04P43	Tenses Part 1
		G11EnU04P44	Tenses Part 2
	➢ Classify adjectives into different types. Change and use degrees of adjectives. ➢ Follow order of adjectives in sentences. ➢ Identify and use adjective phrases and clauses.	G11EnU04P45	Adjectives Part 1
		G11EnU04P46	Adjectives Part 2
		G11EnU04P47	Adjectives Part 3
		G11EnU04P48	Adjective Clause and Adjective Phrase
	➢ Illustrate use of adverbs. ➢ Recognize varying positions of adverbs in sentences according to their kinds and importance. ➢ Identify and use adverbial phrases and clauses	G11EnU04P49	Adverbs
	➢ Illustrate use of prepositions of position, time and movement and direction	G11EnU04P50	Prepositions Part 1
		G11EnU04P51	Prepositions Part 2
	➢ Use in speech and writing, all the appropriate transitional devices	G11EnU04P52	Transitional Devices
	➢ Analyze sentences for different clauses and phrases; evaluate how their positions in	G11EnU04P53	Sentence Structure

	sentences change meaning and affect communicative function.		
	➢ Recognize and use sentence inversion for various purposes.	G11EnU04P54	Sentences and It's Types
	➢ Analyze and construct simple, compound and complex sentences. ➢ Identify, analyze and construct conditional sentences.	G11EnU04P55	Conditionals and Inversions

b. Maths

Unit	SLO	Package Code	Package Name
Number systems	real numbers	G11MU01P01	Number System and Real Numbers
		G11MU01P02	Irrational and Complex Numbers
	Concept of complex numbers and basic operations on them Conjugate and its properties. Modulus (absolute values and its properties	G11MU01P03	Properties and Theorems of Complex Numbers
	Geometrical Representation of Complex numbers by Argand's Diagram.	G11MU01P04	Correspondence and Argand Diagram
Sets functions and groups	Revision of the work done in previous classes	G11MU02P05	Sets and Their Types
		G11MU02P06	Subset and Operation on Set
	Logical Proofs of the Operation on Sets	G11MU02P07	De Morgan's Laws and Logical Statements
		G11MU02P08	Logical Statements
		G11MU02P09	Logical Statements and Distributive Properties
	Functions	G11MU02P10	Relation
		G11MU02P11	Function
	Binary Operations and its Different properties	G11MU02P12	Binary Operations
		G11MU02P13	Properties of Binary Operations
	Groups	G11MU02P14	Abelian Group
		G11MU02P15	Identity Element
		G11MU03P16	Matrix and Its Types

Matrices and Determinants	Revision of the work done in the previous classes	G11MU03P17	Properties of Matrices
	Determinants and their Application in the study of the Algebra of Matrices	G11MU03P18	Determinants, Minors and Cofactors
		G11MU03P19	Adjoint and Inverse of Matrices
	Types of Matrices and the Root and Column Operations on Matrices	G11MU03P20	Elementary Row Operation
		G11MU03P21	Advanced Types of Matrices
		G11MU03P22	Inverse of a Matrix
	Solving Simultaneous Linear System of Equations.	G11MU03P23	System of Equations
Quadratic equations	Revision of the work done in previous classes	G11MU04P24	Quadratic Equations
	Solution of equations Reducible to quadratic equation is one variable	G11MU04P25	Type 1 Reducible Quadratic Equations
		G11MU04P26	Type 2 Reducible Quadratic Equations and Proof of Cube Roots Of Unity
	Cube roots and forth roots of unity	G11MU04P27	Fourth Roots of Unity
	Relation between roots and co-efficient of quadratic equations	G11MU04P28	Nature of Roots and Its Relation with Coefficient
	Formation of quadratic equation from given conditions	G11MU04P29	Formation of Quadratic Equation
	Solution of a system of two equations	G11MU04P30	System of Two Equations Case 1 and 2
		G11MU04P31	System of Two Equations Case 3 and 4
		G11MU04P32	System of Two Equations Case 5
	Problems on quadratic equations		
Partial Fractions	Partial fractions	G11MU05P33	Partial Fractions and Linear Non Repeated Case
		G11MU05P34	Linear Repeated Case
		G11MU05P35	Quadratic Non Repeated Case
		G11MU05P36	Quadratic Repeated Case
Sequence and series	Introduction	G11MU06P37	Sequence and Its Types
	Arithmetic sequence	G11MU06P38	Arithmetic Sequence
	Arithmetic mean/means	G11MU06P39	Arithmetic Mean
	Arithmetic series	G11MU06P40	Arithmetic Series

	Word problems on A.P	G11MU06P41	Word Problems on A.P Part 1
		G11MU06P42	Word Problems on A.P Part 2
	Geometric Progression	G11MU06P43	Geometric Progression
	Geometric mean	G11MU06P44	Geometric Mean
	Geometric series	G11MU06P45	Geometric Series
	Word problems on G.P	G11MU06P46	Word Problems on G.P
	Harmonic Progression	G11MU06P47	Harmonic Progression Part 1
		G11MU06P48	Harmonic Progression Part 2
		G11MU06P49	Harmonic Progression and Sum of First n Natural Numbers
	Sum of the first n Natural Numbers, their squares and cubes		
Permutations, Combinations and probability	Factorial of a Natural Number	G11MU07P50	Factorial of a Natural Number and Permutations
	Permutations	G11MU07P51	Questions Related to Permutations
	Combinations	G11MU07P52	Combinations
	Probability	G11MU07P53	Probability Part 1
		G11MU07P54	Probability Part 2
	Addition and multiplication of probability	G11MU07P55	Multiplication of Probability
Mathematical Induction and Binomial Theorem	Introduction and Application of Mathematical induction	G11MU08P56	Mathematical Induction
	Binomial Theorem	G11MU08P57	Binomial Theorem Part 1
		G11MU08P58	Binomial Theorem Part 2
	Binomial series	G11MU08P59	Binomial Series
Fundamentals of Trigonometry	Introduction	G11MU09P60	Introduction to Trigonometry, Angle Degree and Radians
	Units of measures of angles	G11MU09P61	Units of Measures of Angles
	Relating between the Length of an arc of a circle and circular measure of its central angle		
	Trigonometric functions	G11MU09P62	Trigonometric Functions Part 1
		G11MU09P63	Trigonometric Functions Part 2

Trigonometric Identities of Sum and Difference of Angle	Fundamental formulas of sum and difference of two angles and their properties	G11MU10P64	Distance Formula and Properties of Trigonometric Identities
	Trigonometric ratios of allied angles	G11MU10P65	Fundamental Law of Trigonometry
		G11MU10P66	Proof of Trigonometric Identities Part 1
		G11MU10P67	Proof of Trigonometric Identities Part 2
	Trigonometric ratios of double angles	G11MU10P68	Trigonometric Ratios of Double Angles Part 1
		G11MU10P69	Trigonometric Ratios of Double Angles Part 2
	Sum, difference and product of trigonometric ratios	G11MU10P70	Proof of Sum to Product Formulas
Trigonometric functions and their graphs	Periods of trigonometric functions	G11MU11P71	Periods of Trigonometric Functions
	Graphs of trigonometric functions	G11MU11P72	Graphs of Trigonometric Functions
Application of trigonometry	Heights and distances	G11MU12P73	Heights and Distances
	Cosine formula	G11MU12P74	Proof of Law of Sines and Cosines
	Sine formula		
	Tangent formula	G11MU12P75	Proof Law of Tangents and Sine of Half Angles
	Value of trigonometric ratios of double and half angles of triangles in terms of the sides		
		G11MU12P76	Proof of Cosine and Tangent of Half Angles
	Area of triangular regions	G11MU12P77	Proof of Area of Triangle Case 1 and 2
		G11MU12P78	Proof of Area of Triangle Case 3
	Radii of circles connected with triangles	G11MU12P79	Proof of Circum Radius
		G11MU12P80	Proof of Circum and Ex Radius
		G11MU12P81	Proof of Escribed Radius
		G11MU12P82	Radii of Circles Connected with Triangles
Inverse trigonometric functions	Inverse trigonometric functions	G11MU13P83	Inverse Trigonometric Functions Part 1
		G11MU13P84	Inverse Trigonometric Functions Part 2
		G11MU13P85	Inverse Trigonometric Functions Part 3

Solutions of Trigonometric equations	Solutions of Trigonometric function	G11MU14P86	Trigonometric Functions Part 1
		G11MU14P87	Trigonometric Functions Part 2
		G11MU14P88	Trigonometric Functions Part 3

c. Biology

Unit	SLO	Package Code	Package Titles
CELL STRUCTURE AND FUNCTIONS	List the principles and identify the apparatus used in the techniques of fractionation, differential staining, centrifugation, microdissection, tissue culture, chromatography, electrophoresis and spectrophotometry.	G11BiU01P01	Techniques Used in Cell Biology
	Describe the locations, chemical compositions and significance of the primary and secondary cell walls and of middle lamella.	G11BiU01P02	Cell Wall
	Describe the terms of resolution and magnification with reference to microscopy.	G11BiU01P03	Microscopy, Micrometry and Plasma Membrane
	Explain the use of graticule and micrometer and define the units used in micrometry.		
	Explain the chemical composition of plasma membrane.		
	Rationalize the authenticity of the fluid mosaic model of plasma membrane.		
	Relate the lipid foundation and the variety of proteins of the membrane structure with their roles		
	Identify the role of glycolipids and glycoproteins as the cell surface markers.		
	Explain the role of plasma membrane in regulating cell's interactions with its environment.		
	Describe the chemical nature and metabolic roles of cytoplasm.	G11BiU01P04	Cytoplasm, Endoplasmic Reticulum
	Distinguish between smooth and rough endoplasmic reticulum in terms of their structures and functions.		

	Explain the structure, chemical composition and function of ribosome. Describe the structure and functions of the Golgi complex. State the structure and functions of the Peroxisomes and Glyoxysomes in animal and plant cells.	G11BiU01P05	Ribosomes, Golgi Complex, Peroxisomes and Glyoxysomes
	Describe the formation, structure and functions of the lysosomes. Interpret the storage diseases with reference to the malfunctioning of lysosomes.	G11BiU01P06	Lysosomes
	Explain the external and internal structure of mitochondrion and interlink it with its function.	G11BiU01P07	Mitochondria
	Explain the external and internal structure of chloroplast and interlink it with its function. Describe the structure, composition and functions of centriole.	G11BiU01P08	Chloroplast and Centrosome
	Describe the types, structure, composition and functions of cytoskeleton.	G11BiU01P09	Cytoskeleton
	Describe the chemical composition and structure of nuclear envelope. Compare the chemical composition of nucleoplasm with that of cytoplasm. Explain that nucleoli are the areas where ribosomes are assembled.	G11BiU01P10	Nucleus
	Describe the structure, chemical composition and function of chromosome.	G11BiU01P11	Chromosome
	List the structures missing in prokaryotic cells. Describe the composition of cell wall in a prokaryotic cell. Relate the structure of bacteria as a model prokaryotic cell.	G11BiU01P12	Prokaryotic Cell
	Differentiate between the patterns of cell division in prokaryotic and eukaryotic cells.	G11BiU01P13	Prokaryotic Cell and Eukaryotic Cell
BIOLOGICAL MOLECULES	Introduce biochemistry and describe the approximate chemical composition of protoplasm. Distinguish carbohydrates, proteins, lipids and nucleic acids as the four fundamental kinds of biological molecules.	G11BiU02P01	Biological Molecules in Protoplasm
	Describe and draw sketches of the dehydration-synthesis and hydrolysis reactions for the making and breaking of macromolecule polymers.	G11BiU02P02	Condensation and Hydrolysis

	Explain how the properties of water (high polarity, hydrogen bonding, high specific heat, high heat of vaporization, cohesion, hydrophobic exclusion, ionization and lower density of ice) make it the cradle of life.	G11BiU02P03	Importance of Water
	Define carbohydrates and classify them. Distinguish the properties and roles of monosaccharides, write their empirical formula and classify them.	G11BiU02P04	Carbohydrates and Monosaccharides
	Distinguish the properties and roles of polysaccharides and relate them with the molecular structures of starch, glycogen, cellulose and chitin.	G11BiU02P05	Carbohydrates and Polysaccharides Part 1
		G11BiU02P06	Carbohydrates and Polysaccharides Part 2
	Distinguish the properties and roles of disaccharides and describe glycosidic bond in the transport disaccharides. Compare the isomers and stereoisomers of glucose. Justify that the laboratory-manufactured sweeteners are "left-handed" sugars and cannot be metabolized by the "right-handed" enzymes.	G11BiU02P07	Carbohydrates and Disaccharides
	Define proteins and amino acids and draw the structural formula of amino acid. Outline the synthesis and breakage of peptide linkages.	G11BiU02P08	Proteins and Amino Acids
	List examples and the roles of structural and functional proteins. Justify the significance of the sequence of amino acids through the example of sickle cell hemoglobin. Classify proteins as globular and fibrous proteins.	G11BiU02P09	Proteins
	Define lipids and describe the properties and roles of acylglycerols, phospholipids, terpenes and waxes. Illustrate the molecular structure (making and breaking) of an acylglycerol, a phospholipid and a terpene.	G11BiU02P10	Lipids
	Define nucleic acids and nucleotides. Describe the molecular level structure of nucleotide.	G11BiU02P11	Nucleic Acid

	Distinguish among the nitrogenous bases found in the nucleotides of nucleic acids.		
	Outline the examples of a mononucleotide (ATP) and a dinucleotide (NAD).		
	Explain the general structure of RNA.	G11BiU02P12	RNA
	Distinguish in term of structures and roles, the three types of RNA.		
	Explain the double helical structure of DNA as proposed by Watson and Crick.	G11BiU02P13	DNA
	Illustrate the formation of phosphodiester bond.		
	Define conjugated molecules and describe the roles of common conjugated molecules i.e. glycolipids, glycoproteins, lipoproteins and nucleoproteins.	G11BiU02P16	Molecules
ENZYMES	Describe the structure of enzyme.	G11BiU03P01	Enzyme Structure and Mechanism of Enzyme Action
	Explain how an enzyme catalyzes specific reactions.		
	Define energy of activation and explain through graph how an enzyme speeds up a reaction by lowering the energy of activation.		
	Explain the mechanism of enzyme action through Induced Fit Model, comparing it with Lock and Key Model.		
	Differentiate among the three types of co-factors i.e. in organic ions, prosthetic group and co-enzymes, by giving examples.	G11BiU03P02	Enzymes
	Explain the role and component parts of the active site of an enzyme.		
	Describe the effect of temperature on the rate of enzyme action	G11BiU03P03	Factors Affecting the Rate of Enzyme Action
	Compare the optimum temperatures of enzymes of human and thermophilic bacteria.		
	Describe the range of pH at which human enzymes function		
	Compare the optimum pH of different enzymes like trypsin, pepsin, pepane.		
	Describe how the concentration of enzyme affects the rate of enzyme action.		

	Explain the effect of substrate concentration on the rate of enzyme action.		
	Categorize inhibitors into competitive and non-competitive inhibitors.	G11BiU03P04	Enzyme Inhibition
	Explain feedback inhibition.		
	Describe enzymatic inhibition, its types and its significance.		
	Name the molecules which act as inhibitors.		
	Classify enzymes on the basis of the reactions catalyzed (oxido-reductases, transferases, hydrolases, lyases, isomerases, and ligases).	G11BiU03P05	Classification of Enzymes
	Classify enzymes on the basis of the substrates they use (lipases, amylase, proteases etc).		
BIOENERGETICS	Explain the role of light in photosynthesis.	G11BiU04P01	Photosynthesis
	Identify the two general kinds of photosynthetic pigments (carotenoids and chlorophylls).		
	Describe the roles of photosynthetic pigments in the absorption and conversion of light energy.		
	Differentiate between the absorption spectra of chlorophyll 'a' and 'b'.		
	State the role of CO2 as one of the raw materials of photosynthesis.		
	Explain, narrating the experimental work done, the role of water in photosynthesis.		
	Describe the events of non-cyclic photophosphorylation and outline the cyclic photophosphorylation.	G11BiU04P02	Light Reaction
	Explain the Calvin cycle (the regeneration of RuBP should be understood in outline only).	G11BiU04P03	Dark Reaction
	Explain the process of anaerobic respiration in terms of glycolysis and conversion of pyruvate into lactic acid or ethanol.	G11BiU04P05	Respiration and Breakdown of Pyruvic Acid
	Illustrate the conversion of pyruvate to acetyl-CoA.		
	Outline (naming the reactants and products of each step of) the events of glycolysis.	G11BiU04P06	Glycolysis
	Outline (naming the reactants and products of each step of) the steps of Krebs cycle.	G11BiU04P07	Krebs Cycle

	Describe the arrangement of photosynthetic pigments in the form of photosystem-I and II.	G11BiU04P08	Photosystems and Importance of G3P and Cellular Respiration
	Justify the importance of PGAL in photosynthesis and respiration.		
	Outline the cellular respiration of proteins and fats and correlate these with that of glucose.		
	Define photorespiration and outline the events occurring through it.	G11BiU04P09	Photorespiration
	Outline the process of C4 photosynthesis as an adaptation evolved in some plants to deal with the problem of photorespiration.		
	Rationalize how the disadvantageous process of photorespiration evolved.		
	Explain the effect of temperature on the oxidative activity of RuBP carboxylase.		
ACELLULAR LIFE	Justify the status of viruses among living and non-living things.	G11BiU05P01	Discovery and Structure of Virus
	Trace the history of viruses since their discovery.		
	Explain the structure of a model bacteriophage, flu virus and HIV.		
	Outline the usage of bacteriophage in genetic engineering.		
	Describe the Lytic and Lysogenic life cycles of a virus.	G11BiU05P02	Life Cycle of Bacteriophage
	Justify why a virus must have a host cell to parasitize in order to complete its life cycle.	G11BiU05P03	Parasitic Nature of Virus
	Explain how a virus survives inside a host cell, protected from the immune system.		
	Determine the method a virus employs to survive/ pass over unfavorable conditions when it does not have a host to complete the life cycle.		
	Classify viruses on the bases of their hosts and structure.		
	Justify the name of the virus i.e., "Human Immunodeficiency Virus" by establishing T-helper cells as the basis of immune system.	G11BiU05P05	Acquired Immunodeficiency Syndrome
	List the symptoms of AIDS.		
	Describe the treatments available for AIDS.		
	List some common control measures against the transmission of HIV.		

	Describe the causative agent, symptoms, treatment and prevention of the following viral diseases: hepatitis, herpes, polio and leaf curl virus disease of cotton.	G11BiU05P06	Viral Diseases
	List the sources of transmission for each of the above-mentioned diseases.		
	Assess from the given data the economic loss from viral infections (cotton leaf curl virus disease and bird flu virus) in Pakistan.		
	Explain the life cycle of HIV.	G11BiU05P07	HIV, Prions and Viroids
	Reason out the specificity of HIV on its host cells.		
	Describe the structure of prions and viroids.		
	List the diseases caused by prions and viroids.		
PROKARYOTES	Outline the taxonomic position of prokaryotes in terms of domains archaea and bacteria and in terms of kingdom monera.	G11BiU06P01	Taxonomy of Prokaryotes and Archaea
	Explain the phylogenetic position of prokaryotes.		
	List the unifying archeal features that distinguish them from bacteria.		
	Explain that most Archaea inhabit extreme environments.		
	Explain the great diversity of shapes and sizes found in bacteria.	G11BiU06P04	Diversity of Shapes and Sizes in Bacteria
	Describe autotrophic and heterotrophic nutrition in bacteria.	G11BiU06P05	Modes of Nutrition in Bacteria
	Explain motility in bacteria.	G11BiU06P06	Motility and Growth in Bacteria
	List the phases in the growth of bacteria.		
	Describe different methods of reproduction in bacteria.	G11BiU06P09	Reproduction in Bacteria
	Justify the endospore formation in bacteria to withstand unfavorable conditions.		
	Describe bacteria as recyclers of nature.	G11BiU06P11	Importance of Bacteria
	Outline the ecological and economic importance of bacteria.		
	Explain the use of bacteria in research and technology.		
	List the chemical and physical methods used to control harmful bacteria.		
	Describe important bacterial diseases in man e.g. cholera, typhoid, tuberculosis, and		

	pneumonia; emphasizing their symptoms, causative bacteria, treatments, and preventative measures.		
PROTISTS AND FUNGI	Describe the salient features with examples of protozoa, algae, myxomycota and oomycota as the major groups of protists.	G11BiU07P02	Major Groups of Protists
	Explain protists as a diverse group of eukaryotes that has polyphyletic origin and defined only by exclusion from other groups.	G11BiU07P03	Protists and Fungi
	List the characteristics that distinguish fungi from other groups and give reasons why fungi are classified in a separate kingdom.		
	Explain yeast as unicellular fungi that are used for baking and brewing and are also becoming very important for genetic research.		
	Give examples of edible fungi.		
	Name a few fungi from which antibiotics are obtained.		
	Classify fungi into zygomycota, ascomycota and basidiomycota and give the diagnostic features of each group.	G11BiU07P04	Mutualistic Association and Classification of Fungi
	Explain the mutualism established in mycorrhizae and lichen associations.		
	Describe the ecological impact of fungi causing decomposition and recycling of materials.	G11BiU07P05	Ecological Impact of Fungi
	Explain the pathogenic role of fungi.		
DIVERSITY AMONG PLANTS	Outline the evolutionary origin of plants.	G11BiU08P01	The Evolutionary Origin of Plants
	List the diagnostic features shared by all plants, with emphasis on alternation of generation.		
	Describe the general characteristics of bryophytes.	G11BiU08P02	General Characteristics and Land Adaptations of Bryophytes
	Explain the land adaptations of bryophytes.		
	Outline the life cycle of moss.	G11BiU08P03	Life Cycle of Moss
	Describe the general characteristics of vascular plants.	G11BiU08P05	Tracheophytes
	List the characters of seedless vascular plants with examples of whisk ferns, club mosses, horsetails and ferns.	G11BiU08P06	Major Groups of Vascular Plants
	Summarize the importance of seedless vascular plants.		
	Outline the life cycle of ferns.	G11BiU08P07	Life Cycle of Fern

	Describe the evolution of seed.	G11BiU08P08	Evolution of Seed
	Explain the life cycle of a flowering plant.	G11BiU08P09	Life Cycle of Flowering Plants
	Explain how this life cycle demonstrates an adaptation of angiosperms on land.	G11BiU08P09	
	Define angiosperms and explain the difference between monocots and dicots.	G11BiU08P10	Angiosperms and Evolution of Leaf
	Explain the evolution of leaf in vascular plants.		
	Define inflorescence and describe its major types.	G11BiU08P11	Inflorescence
DIVERSITY AMONG ANIMALS	Describe the general characteristics of animals.	G11BiU09P01	Characteristics and Criteria for Animal Classification
	Classify animals on the base of presence and absence of tissues.		
	Differentiate the diploblastic and triploblastic levels of organization.		
	Describe the types of symmetry found in animals.	G11BiU09P02	Criteria for Animal Classification
	Differentiate pseduocoelomates, acoelomates and coelomates.		
	Classify coelomates into protostomes and deuterostomes.		
	Describe the general characteristics, importance and examples of sponges, cnidarians, platyhelminths, aschelminths (nematodes), mollusks, annelids, arthropods and echinoderms.	G11BiU09P03	Phylum Porifera
		G11BiU09P04	Phylum Cnidaria
		G11BiU09P05	Phylum Platyhelminthes and Phylum Aschelminthes
	Describe the evolutionary adaptations in the concerned phyla for digestion, gas exchange, transport, excretion, and coordination.	G11BiU09P06	Phylum Annelida
		G11BiU09P07	Phylum Mollusca
		G11BiU09P08	Phylum Arthropoda
	List the diagnostic characteristics of jawless fishes, cartilaginous fishes and bony fishes.	G11BiU09P09	Phylum Chordata and Craniata
	Describe the general characteristics of amphibians, reptiles, birds and mammals.	G11BiU09P10	Class Amphibia
		G11BiU09P11	Class Reptilia
	Differentiate among monotremes, marsupials, and placentals.	G11BiU09P12	Class Aves and Class Mammalia
	Describe the evolutionary adaptations in concerned groups for		

	gas exchange, transport and coordination.		
FORM AND FUNCTIONS IN PLANTS	Explain movement of water into or out of cell in isotonic, hypotonic, and hypertonic conditions.	G11BiU10P09	Homeostasis and Balance of Water and Solutes
	Describe osmotic adjustments in hydrophytic (marine and freshwater), xerophytic and mesophytic plants.	G11BiU10P10	Homeostasis and Osmoregulation in Plants
	Explain the osmotic adjustments of plants in saline soils.		
	List the adaptations in plants to cope with low and high temperatures.	G11BiU10P11	Homeostasis and Thermoregulation in Plants
	Describe the structure of supporting tissues in plants.	G11BiU10P12	Support in Plants
	Explain the turgor pressure and explain its significance in providing support to herbaceous plants.		
	Define growth and explain primary and secondary growth in plants.	G11BiU10P13	Growth and Development in Plants
	Describe the role of apical meristem and lateral meristem in primary and secondary growth.		
	Explain the types of movement in plants in response to light, force of gravity, touch and chemicals.	G11BiU10P17	Movement in Plants
DIGESTION	Describe the mechanical and chemical digestion in oral cavity.	G11BiU11P01	Digestive System of Man, Buccal Cavity and Pharynx
	Explain swallowing and peristalsis.		
	Describe the structure of stomach and relate each component with the mechanical and chemical digestion in stomach.	G11BiU11P02	Digestive System of Man and Stomach
	Explain the role of nervous system and gastrin hormone on the secretion of gastric juice.		
	Describe the major actions carried out on food in the three regions of the small intestine.	G11BiU11P03	Digestive System of Man and Small Intestine
	Relate the secretion of bile and pancreatic juice with the secretin hormone		
	Explain the absorption of digested products from the small intestine lumen to the blood capillaries and lacteals of the villi.		
	Describe composition of bile and relate the constituents with respective roles.		
	Describe the component parts of large intestine with their respective roles.	G11BiU11P04	Digestive System of Man, Large Intestine and Glands

	Correlate the involuntary reflex for egestion in infants and the voluntary control in adults.		
	Explain the storage and metabolic role of liver.		
	Outline the structure of pancreas and explain its function as an exocrine gland.		
	Describe the causes, prevention, and treatment of the following disorders; ulcer, food poisoning, dyspepsia.	G11BiU11P05	Disorders of Digestive System and Food Habits Part 1
	Describe obesity in terms of its causes, preventions and related disorders.	G11BiU11P06	Disorders of Digestive System and Food Habits Part 2
	Explain the symptoms and treatments of bulimia nervosa and anorexia nervosa.		
CIRCULATION	State the phases of heartbeat.	G11BiU12P01	Heartbeat and Its Control
	Describe the flow of blood through heart as regulated by the valves.		
	Explain the role of SA node, AV node and Purkinji fibers in controlling the heartbeat.		
	List the principles and uses of Electrocardiogram.		
	Describe the structure of the walls of heart and rationalize the thickness of the walls of each chamber.	G11BiU12P02	Heart and Vascular Pathway
	Trace the path of the blood through the pulmonary and systemic circulation (coronary, hepatic-portal and renal circulation).		
	State the location of heart in the body and define the role of pericardium.		
	Define blood pressure and explain its periods of systolic and diastolic pressure.	G11BiU12P04	Blood Pressure
	Describe the detailed structure of arteries, veins and capillaries.	G11BiU12P05	Blood Vessels
	Describe the role of arterioles in vasoconstriction and vasodilation.	G11BiU12P06	Role of Arterioles and Role of Precapillary Sphincters
	Describe the role of precapillary sphincters in regulating the flow of blood through capillaries.		
	Define the term thrombus and differentiate between thrombus and embolus.	G11BiU12P07	Cardiovascular Diseases
	Identify the factors causing atherosclerosis and arteriosclerosis.		
	Categorize Angina pectoris, heart attack, and heart failure as the		

	stages of cardiovascular disease development.		
	State the congenital heart problem related to the malfunctioning of cardiac valves.		
	Describe the principles of angiography.	G11BiU12P08	Treatment and Prevention of Cardiovascular Diseases
	Outline the main principles of coronary bypass, angioplasty and open-heart surgery.		
	Define hypertension and describe the factors that regulate blood pressure and can lead to hypertension and hypotension.	G11BiU12P09	Hypertension and Hypotension
	List the changes in life styles that can protect man from hypertension and cardiac problems.		
	Describe the formation, composition and function of interstitial fluid.	G11BiU12P10	Lymphatic System
	Compare the composition of interstitial fluid with that of lymph.		
	State the structure and role of lymph capillaries, lymph vessels and lymph trunks.		
	Describe the role of lymph vessels (lacteals) present in villi.		
	Describe the functions of lymph nodes and state the role of spleen as containing lymphoid tissue.		
IMMUNITY	Describe the structural features of human skin that make it impenetrable barrier against invasion by microbes.	G11BiU13P01	First Line of Defence
	Explain how oil and sweat glands within the epidermis inhibit the growth and also kill microorganisms.		
	Recognize the role of the acids and enzymes of the digestive tract in killing the bacteria present in food.		
	State the role of the ciliated epithelium of nasal cavity and of the mucous of the bronchi and bronchioles in trapping air borne microorganisms.		
	State how the proteins of the complement system kill bacteria and how the interferons inhibit the ability of viruses to infect cells.	G11BiU13P02	Second Line of Defence
	State the events of the inflammatory response as one of the most generalized nonspecific defenses.		

	Outline the release of pyrogens by microbes and their effect on hypothalamus to boost the body's temperature.		
	List the ways the fever kills microbes.		
	Categorize the immune system that provides specific defense and acts as the most powerful means of resisting infection.	G11BiU13P03	Third Line of Defence
	State the inborn and acquired immunity as the two basic types of immunity.	G11BiU13P04	Immune System
	Differentiate the two types of acquired immunity (active and passive immunity).		
	Identify the process of vaccination as a means to develop active acquired immunity.		
	Identify monocytes, T-cells and B-cells as the components of the immune system.		
	Explain the role of memory cells in long-term immunity.		
	Describe the role of macrophages and neutrophils in killing bacteria.		
	Explain how the Natural Killer (NK) cells kill the cells that are infected by microbes and also kill cancer cells.		
	Describe the roles T-cells in cell-mediated immunity.	G11BiU13P05	Role of Cells in Immunity
	Describe the role of B-cells in antibody-mediated immunity.		
	Draw the structural model of an antibody molecule.	G11BiU13P06	Structure of Antibody
	Define allergies and correlate the symptoms of allergies with the release of histamines.	G11BiU13P07	Disorders of the Immune System
	Describe the autoimmune diseases.		
	Describe the role of T-cells and B-cells in transplant rejections.		
RESPIRATION	Describe the main structural features and functions of the components of human respiratory system.	G11BiU14P01	Respiratory System of Man
	Describe the ventilation mechanism in humans.	G11BiU14P02	The Mechanism of Breathing and Respiratory Volumes
	State lung volumes and capacities.		
	Describe the transport of oxygen and carbon dioxide through blood.	G11BiU14P04	Transport of Gases
	Describe the role of respiratory pigments.		

PROKARYOTES	Justify the occurrence of bacteria in the widest range of habitats.	G11BiU06P02	Bacteria
	List the diagnostic features of the major groups of bacteria.		
	Describe detailed structure and chemical composition of bacterial cell wall and other coverings.		
	Compare cell wall differences in Gram-positive and Gram-negative bacteria.		

d. Chemistry

Unit	SLO	Package Code	Package Titles
Stoichiometry	Interpret a balanced chemical equation in terms of interacting moles, representative particles, masses and volumes of gases (at STP). (Analyzing)	G11ChU01P01	Mole Ratio
	Construct mole ratios from balanced equations for use as conversion factors in stoichiometric problems. (Applying)		
	Perform stoichiometric calculations with balanced equations using represntative particles. (analyzing)		
	Perform stoichiometric calculations with balanced equations using moles, masses. (analyzing)	G11ChU01P02	Stoichiometry, Mass-Mass Relationship Part 1
	Perform stoichiometric calculations with balanced equations using masses, volume. (analyzing)	G11ChU01P03	Stoichiometry, Mass-Mass Relationship Part 2

	Identify the limiting reagent in a reaction. (Analyzing)	G11ChU01P04	Limiting Reactant
	Given the information from which any two of the following may be determined, calculate the theoretical yield, actual yield, and percentage yield. (Understanding) Calculate the theoretical yield and the percent yield when given the balanced equation, the amounts of reactants and the actual yield. (Applying)	G11ChU01P05	Yield
Atomic Structure	Summarize Bohr's atomic theory (Applying) Use Bohr's model for calculating radii of orbits. (Understanding)	G11ChU02P01	Bohr's Atomic Model
	Use Bohr's atomic model for calculating energy of electron in a given orbit of hydrogen atom. Relate energy equation (for electron) to frequency, wave length and wave number of radiation emitted or absorbed by the electron.	G11ChU02P02	Derivation of the Energy of Revolving Electron in nth Orbit
		G11ChU02P03	Derivation of Energy Difference, Wavenumber, and Frequency
	Describe the concept of orbitals. (Understanding) Distinguish among principal energy levels, energy sub levels, and atomic	G11ChU02P05	Orbital and Quantum Numbers

	orbitals. (Understanding)		
	Describe the general shapes of s, p, and d orbitals. (Understanding)	G11ChU02P06	Shape of Orbitals
	Relate the discrete-line spectrum of hydrogen to energy levels of electrons in the hydrogen atom. (Applying)	G11ChU02P07	Spectrum and Hydrogen Spectrum
	Describe the orbitals of the hydrogen atom in order of increasing energy. (Understanding)		
	Use the Aufbau Principle, the Pauli Exclusion Principle, and Hund's Rule to write the electronic configuration of the elements. (Applying)	G11ChU02P08	Electronic Configuration
	Explain the sequence of filling of electrons in many electron atoms. (Applying)	G11ChU02P09	Writing Electronic Configuration
Theories of Covalent Bonding and Shapes of Molecules	Use VSEPR and VBT theories to describe the shapes of simple covalent molecules. (Applying)	G11ChU03P01	VSEPR Theory and Hybrid Orbital Theory
		G11ChU03P02	Valence Bond Theory
	Describe the features of sigma and pi bonds. (Understanding)		
	Describe the shapes of simple molecules using orbital hybridization. (Applying)	G11ChU03P03	sp3 Hybridization
		G11ChU03P04	sp2 and sp Hybridization
	Determine the shapes of some molecules from the	G11ChU03P05	Shape of Molecules Part 1

	number of bonded pairs and lone pairs of electrons around the central atom. (Analyzing)	G11ChU03P06	Shape of Molecules Part 2
	Define bond energies and explain how they can be used to compare bond strengths of different chemical bonds. (Analyzing)	G11ChU03P07	Bond Energy
	Predict the molecular polarity from the shapes of molecules. (Applying) Describe how knowledge of molecular polarity can be used to explain some physical and chemical properties of molecules. (Analyzing)	G11ChU03P08	Dipole Moment
	Describe the change in bond lengths of hetero-nuclear molecules due to differences in the Electronegativity values of bonded atoms. (Understanding) Explain what is meant by the term ionic character of a covalent bond. (Understanding)	G11ChU03P09	Bond Length
States of Matter I: Gases	List the postulates of Kinetic Molecular Theory. (Remembering) Describe the motion of particles of a gas according to Kinetic Theory. (Applying)	G11ChU04P01	Kinetic Molecular Theory of Gaseous State

	Use Kinetic Theory to explain gas pressure. (Applying)		
	Relate temperature to the average kinetic energy of the particles in a substance. (Applying)		
	Describe the effect of change in pressure on the volume of gas. (Applying)	G11ChU04P02	Boyle's Law
	Describe the effect of change in temperature on the volume of gas. (Applying)	G11ChU04P03	Charles's Law
	Explain the significance of absolute zero, giving its value in degree Celsius and Kelvin. (Understanding)	G11ChU04P04	Absolute Zero
	State and explain the significance of Avogadro's Law. (Understanding) State the values of standard temperature and pressure (STP). (Remembering)	G11ChU04P05	Avogadro’s Law and General Gas Equation
	Derive Ideal Gas Equation using Boyle's, Charles' and Avogadro's law. (Understanding)		
	Explain the significance and different units of ideal gas constant. (Understanding)	G11ChU04P06	General Gas Constant
	Distinguish between real and ideal gases. (Understanding)	G11ChU04P07	Behavior of Real and Ideal Gases
	Explain why real gases deviate from the gas laws. (Analyzing)		

	Derive new form of Gas Equation with volume and pressure corrections for real gases. (Understanding)		
	State and use Graham's Law of Diffusion. (Understanding) Describe some of the implications of the Kinetic Molecular Theory, such as the velocity of molecules and Graham's Law. (Applying)	G11ChU04P08	Graham's Law of Diffusion
	State and use Dalton's Law of Partial Pressures. (Understanding)	G11ChU04P09	Dalton's Law of Partial Pressures
	Explain Lind's method for the liquefaction of gases. (Understanding)	G11ChU04P10	Liquefaction of Gases
	Define pressure and give its various units. (Remembering)		
	Define and explain plasma formation. (Understanding)	G11ChU04P11	Plasma State
	Define and describe the properties of Plasma.(Applying)		
States of Matter II: Liquids	Describe simple properties of liquids e.g., diffusion, compression, expansion, motion of molecules, spaces between them, intermolecular forces and kinetic energy based on Kinetic Molecular Theory. (Understanding)	G11ChU05P01	Liquid State
	Explain applications of dipole-dipole forces, hydrogen	G11ChU05P02	Application of Dipole-Dipole, London Force, Hydrogen Bonding

	bonding and London forces. (Applying)		
	Explain applications of dipole-dipole forces, hydrogen bonding and London forces. (Applying) Use the concept of Hydrogen bonding to explain the following properties of water: high surface tension, high specific heat, low vapor pressure, high heat of vaporization, and high boiling point. And anomalous behaviour of water when its density shows a maximum at 4-degree centigrade(Applying)	G11ChU05P06	Hydrogen Bonding and Its Properties
	Define molar heat of fusion and molar heat of vaporization. (Remembering) Describe how heat of fusion and heat of vaporization affect the particles that make up matter. (Understanding)	G11ChU05P07	Energetics of Phase Change
	Relate energy changes with changes in intermolecular forces. (Applying)		
	Define dynamic equilibrium between two physical states. (Remembering)	G11ChU05P08	Liquid Crystal
	Describe liquid crystals and give their uses in daily life. (Applying)		
	Differentiate liquid crystals from pure		

	liquids and crystalline solids. (Applying)		
States of Matter III: Solids	Describe simple properties of solids e.g., diffusion, compression, expansion, motion of molecules, spaces between them, intermolecular forces and kinetic energy based on kinetic molecular theory. (Understanding)	G11ChU06P01	Solid State
	isomorphism, polymorphism, allotropy and transition temperature. (Understanding)		
	Differentiate between amorphous and crystalline solids. (Understanding) Describe properties of crystalline solids like geometrical shape, melting point, cleavage planes, habit of a crystal, crystal growth, anisotropy, symmetry,	G11ChU06P02	Classification of Solids
	Explain the significance of the unit cell to the shape of the crystal using NaCl as an example. (Applying)	G11ChU06P04	Structure of NaCl
	Define lattice energy. (Remembering)		
	Name three types of packing arrangements and draw or construct models of them. (Applying)	G11ChU06P05	Structure of Metals

	Name three factors that affect the shape of an ionic crystal. (Understanding) Differentiate between ionic, covalent, molecular and metallic crystalline solids. (Applying)	G11ChU06P06	Metallic and Ionic Crystals
		G11ChU06P07	Molecular and Covalent Crystals
	Define and explain molecular and metallic solids. (Understanding)		
Chemical Equilibrium	Define chemical equilibrium in terms of a reversible reaction. (Remembering) State the necessary conditions for equilibrium and the ways that equilibrium can be recognized. (Understanding)	G11ChU07P01	Chemical Equilibrium
	Write the equilibrium expression for a given chemical reaction. (Understanding) Write expression for reaction quotient.		
	Determine if the reactants or products are favored in a chemical reaction, given the equilibrium constant. (Analyzing)	G11ChU07P03	Application of Law of Equilibrium
	State Le Chatelier's Principle and be able to apply it to systems in equilibrium with changes in concentration, pressure, temperature, or the addition of catalyst.	G11ChU07P04	Le Chatelier's Principle

	(Applying) Determine if the equilibrium constant will increase or decrease when the temperature is changed, given the equation for the reaction. (Applying)		
	Explain industrial applications of Le Chatelier's Principle using Haber's process as an example. (Analyzing)		
	Define and explain solubility product. (Understanding)	G11ChU07P05	Solubility Product and Common Ion Effect
	Define and explain common ion effect giving suitable examples. (Applying)		
Acids, Bases and Salts	Define Bronsted and Lowery concepts for acids and bases(Remembering)	G11ChU08P01	Bronsted and Lowery's Concepts for Acids and Bases
	Identify conjugate acid-base pairs of Bronsted-Lowery acid and base(Analyzing)		
	Define salts, conjugate acids and conjugate bases. (Remembering)		
	Explain ionization constant of water and calculate pH and pOH in aqueous medium using given Kw values. (Applying)	G11ChU08P02	Ionic Product of Water, Acids, and Bases
	Use the extent of ionization and the acid dissociation constant, Ka, to distinguish between strong and weak acids. (Applying)		
	Use the extent of ionization and the		

	base dissociation constant, Kb, to distinguish between strong and weak bases. (Applying)		
	Define a buffer, and show with equations how a buffer system works. (Applying)	G11ChU08P03	Buffer Solution
	Make a buffered solution and explain how such a solution maintains a constant pH, even with the addition of small amounts of strong acid or strong base. (Understanding)	G11ChU08P04	Making Buffer Solution
	Use the concept of hydrolysis to explain why aqueous solutions of some salts are acidic or basic. (Applying)	G11ChU08P05	Hydrolysis and Levelling Effect
	Use concept of hydrolysis to explain why the solution of a salt is not necessarily neutral. (Understanding)		
	Define and explain leveling effect. (Understanding)		
Chemical Kinetics	Define chemical kinetics. (Remembering)	G11ChU09P01	Chemical Kinetics
	Given a potential energy diagram for a reaction, discuss the reaction mechanism for the reaction. (Applying)		
	Explain and use the terms rate of reaction, rate equation, order of reaction, rate constant, and rate-determining step. (Understanding)	G11ChU09P02	Rate of Reaction

	Explain qualitatively factors affecting rate of reaction. (Applying)	G11ChU09P03	Factors Affecting Rate of Reaction Part 1
	Explain the effects of concentration, temperature, and surface area on reaction rates. (Applying)	G11ChU09P04	Factors Affecting Rate of Reaction Part 2
	Use the collision theory to explain how the rate of a chemical reaction is influenced by the temperature, concentration, size of molecules and. (Applying)		
	Describe that increase in collision energy by increasing the temperature can improve the collision frequency. (Applying)		
	Given the order with respect to each reactant, write the rate law for the reaction. (Applying)	G11ChU09P05	Writing Rate Law
	Explain what is meant by the terms activation energy and activated complex. (Understanding) Relate the ideas of activation energy and the activated complex to the rate of a reaction. (Applying)	G11ChU09P06	Activation Energy
	Explain the significance of the rate-determining step on the overall rate of a multi step reaction. (Analyzing)		

	Describe the role of the rate constant in the theoretical determination of reaction rate. (Applying)	G11ChU09P07	Determination of Rate of Reaction
	Define terms catalyst, catalysis, homogeneous catalysis and heterogeneous catalysis. (Understanding)	G11ChU09P08	Catalysis and Its Types
	Explain that a catalyst provides a reaction pathway that has low activation energy. (Applying)	G11ChU09P09	Characteristics of Catalysts
	Describe enzymes as biological catalysts. (Understanding)	G11ChU09P10	Enzyme Catalysts
Solutions and Colloids	Describe types of colloids and their properties. (Applying)	G11ChU10P02	Colloids
	Define hydrophilic and hydrophobic molecules. (Remembering)	G11ChU10P03	Hydrophobic and Hydrophilic Molecules
	Define the term water of hydration. (Remembering)		
	Explain the effect of temperature on solubility and interpret the solubility graph. (Analyzing)	G11ChU10P04	Solubility
	Distinguish between the salvation of ionic species and molecular substances. (Understanding)		
	List three factors that accelerate the dissolution process. (Understanding)		

	Explain concept of solubility and how it applies to solution saturation. (Applying)		
	Express solution concentration in terms of mass percent, molality, molarity, parts per million, billion and trillion, and mole fraction. (Remembering)	G11ChU10P05	Concentration Units of Solution Part 1
		G11ChU10P06	Concentration Units of Solution Part 2
		G11ChU10P07	Concentration Units of Solution Part 3
	Explain the nature of solutions in liquid phase giving examples of completely miscible, partially miscible and immiscible liquid-liquid solutions. (Applying)	G11ChU10P08	Solution of Liquid in Liquid
	Describe the role of salvation in the dissolving process. (Understanding)	G11ChU10P09	Energetics of Solution
	Define heat of solution and apply this concept to the hydration of ammonium nitrate crystals. (Applying)		
	Define the terms colligative. (Remembering)	G11ChU10P11	Colligative Properties
	List some colligative properties of liquids. (Understanding)		
	Explain how solute particles may alter the colligative properties. (Applying)		
	Explain on a particle basis how the addition of a solute to a pure solvent causes an elevation of the boiling point	G11ChU10P12	Elevation of Boiling Point and Its Measurement
		G11ChU10P13	Elevation of Boiling Point and Depression of Freezing Point

	and depression of the freezing point of the resultant solution. (Applying)		
	Describe on a particle basis why a solution has a lower vapor pressure than the pure solvent. (Applying)	G11ChU10P14	Elevation of Boiling Point
Thermochemistry	Define thermodynamics. (Remembering)	G11ChU11P01	Thermodynamics
	Classify reactions as exothermic or endothermic. (Understanding)		
	Define the terms system, surrounding, boundary, state function, heat, heat capacity, internal energy, work done and enthalpy of a substance. (Remembering)		
	Specify conditions for the standard heat of reaction. (Applying)	G11ChU11P05	Enthalpy
	Describe how heat of combustion can be used to estimate the energy available from foods. (Analyzing)	G11ChU11P06	Measurement of Enthalpy of Reaction
	Apply Hess's Law to construct simple energy cycles. (Understanding)		
	Explain reaction pathway diagram in terms of enthalpy changes of the reaction. (Born Haber's Cycle) (Applying)	G11ChU11P07	Hess’s Law and the Born Haber Cycle
Electrochemistry	Give the characteristics of a	G11ChU12P01	Oxidation and Reduction

	Redox reaction. (Understanding)		
	Define oxidation and reduction in terms of a change in oxidation number. (Applying)		
	Determine the oxidation number of an atom of any element in a pure substance. (Applying)	G11ChU12P02	Finding Oxidation Number
	Use the oxidation-number change method to identify atoms being oxidized or reduced in redox reactions. (Applying) Use the oxidation-number change method to balance redox equations. (Applying)	G11ChU12P03	Redox Reaction, Ion Electron, Half-Reaction Method Part 1
	Balance redox reactions that take place in acid solutions. (Applying) Break a redox reaction into oxidation and reduction half reactions. (Applying)	G11ChU12P04	Redox Reaction, Ion Electron, Half-Reaction Method Part 2
	When given an unbalanced redox equation, use the half reaction method to balance the equation. (Applying)		
	Explain how a fuel cell produces electrical energy. (Applying)	G11ChU12P10	Fuel Cell
	State and explain Faraday's laws. (Understanding)	G11ChU12P11	Faraday's Law
	Describe how a dry cell supplies electricity. (Understanding)	G11ChU12P12	Alkaline Batteries, Dry Cell

	Identify the substance oxidized and the substance reduced in a dry cell. (Applying)		
Atomic Structure	Explain production, properties, types and uses of X-rays. (Understanding) Define photon as a unit of radiation energy. (Remembering)	G11ChU02P04	X-Ray and Planck's Quantum Theory
States of Matter II: Liquids	Explain physical properties of liquids such as evaporation, vapour pressure, boiling point, viscosity, and surface tension. (Understanding)	G11ChU05P03	Vapour Pressure
	Explain physical properties of liquids such as evaporation, vapour pressure, boiling point, viscosity, and surface tension. (Understanding)	G11ChU05P04	Evaporation and Boiling Point
	Explain physical properties of liquids such as evaporation, vapour pressure, boiling point, viscosity, and surface tension. (Understanding)	G11ChU05P05	Viscosity and Surface Tension
Thermochemistry	Name and define the units of thermal energy. (Remembering) Define bond dissociation energy. (Remembering)	G11ChU11P02	Units of Thermal Energy and Bond Dissociation Energy

Electrochemistry	Define cathode, anode, electrode potential and S.H.E. (Standard Hydrogen Electrode). (Remembering) Define the standard electrode potential of an electrode. (Remembering)	G11ChU12P05	Electrode Potential
	Define cell potential, and describe how it is determined. (Understanding)		
	Use the activity series of metals to predict the products of single replacement reactions. (Analysis)	G11ChU12P06	Reactivity Series
	Describe the reaction that occurs when a lead storage battery is recharged. (Applying)	G11ChU12P07	Lead Storage Battery
	Explain how a lead storage battery produces electricity. (Understanding)		

e. Physics

Unit	SLO	Package Code	Package Title
Measurement	describe the scope of Physics in science, technology and society.	G11PhU01P01	Introduction to Physics
	explain why all measurements contain some uncertainty.	G11PhU01P03	Errors and Uncertainty
	distinguish between systematic errors (including zero errors) and random errors.		
	differentiate between precision and accuracy.	G11PhU01P05	Precision and Accuracy
	assess the uncertainty in a derived quantity by simple addition of actual, fractional or percentage uncertainties.	G11PhU01P06	Further Discussion on Error and Uncertainties
	check the homogeneity of physical equations by using dimensionality and base units	G11PhU01P08	Dimensions
	derive formulae in simple cases using dimensions.		
Vectors and Equilibrium	determine the sum of vectors using head to tail rule.	G11PhU02P02	Head to Tail Rule
	represent a vector into two perpendicular components.	G11PhU02P03	Rectangular Components of Vector
	determine the sum of vectors using perpendicular components.	G11PhU02P04	Addition of Vectors by Rectangular Components
	state first condition of equilibrium.	G11PhU02P10	Equilibrium and Its Condition
	state second condition of equilibrium.		
Measurement	state SI base units, derived units, and supplementary units for various measurements.	G11PhU01P02	SI Base Units, Derived Units, and Their Conventions Part 1
	express derived units as products or quotients of the base units.		
	state the conventions for indicating units as set out in the SI units.	G11PhU01P02A	SI Base Units, Derived Units, and Their Conventions Part 2
	quote answers with correct scientific notation, number of	G11PhU01P07	Significant Figures and Its Rules

	significant figures and units in all numerical and practical work.	G11PhU01P07A	Rules of Significant Figures and Scientific Notations
Vectors and Equilibrium	describe the Cartesian coordinate system.	G11PhU02P01	Cartesian Coordinate System
	describe vector product of two vectors in term of angle between them.	G11PhU02P06	Vector Product
	state the method to determine the direction of vector product of two vectors.	G11PhU02P07	Calculating Vector Product Along Its Direction
	define the torque as vector product r x F.	G11PhU02P08	Torque
	list applications of torque or moment due to a force.	G11PhU02P09	Applications of Torque
	describe scalar product of two vectors in term of angle between them.	G11PhU02P05	Scalar Product
	solve two dimensional problems involving forces (statics) using 1st and 2nd conditions of equilibrium.	G11PhU02P11	Problems Related to Equilibrium
Forces and Motion	describe vector nature of displacement.	G11PhU03P01	Displacement, Speed, Velocity and Acceleration
	describe average and instantaneous velocities of objects.		
	compare average and instantaneous speeds with average and instantaneous velocities.		
	define average acceleration (as rate of change of velocity aav = Δv / Δt) and instantaneous acceleration (as the limiting value of average acceleration when time interval Δt approaches zero).		
	distinguish between positive and negative acceleration, uniform and variable acceleration.		
	interpret displacement-time and velocity-time graphs of objects moving along the same straight line.	G11PhU03P02	Velocity and Acceleration From Graphs
	determine the instantaneous velocity of an object moving along the same straight line by measuring the slope of displacement-time graph.		
	determine the instantaneous acceleration of an object measuring the slope of velocity-time graph.		
	explain that projectile motion is two dimensional motion in a vertical plane.	G11PhU03P03	Projectile Motion

	communicate the ideas of a projectile in the absence of air resistance that. (i) Horizontal component (VH) of velocity is constant. (ii) Acceleration is in the vertical direction and is the same as that of a vertically free falling object. (iii) The horizontal motion and vertical motion are independent of each other.		
	evaluate using equations of uniformly accelerated motion that for a given initial velocity of frictionless projectile. 1. How higher does it go? 2. How far would it go along the level land? 3. Where would it be after a given time? 4. How long will it remain in air?	G11PhU03P04	Projectile Motion (Maximum Height and Total Time of Flight)
	determine for a projectile launched from ground height. 1. launch angle that results in the maximum range. 2. relation between the launch angles that result in the same range.	G11PhU03P05	Projectile Motion (Range)
	apply Newton's laws to explain the motion of objects in a variety of context.	G11PhU03P07	Newton's Law of Motion and Momentum
	define mass (as the property of a body which resists change in motion).		
	describe the Newton's second law of motion as rate of change of momentum.		
	show awareness that Newton's Laws are not exact but provide a good approximation, unless an object is moving close to the speed of light or is small enough that quantum effects become significant.		
	define Impulse (as a product of impulsive force and time).		
	describe the effect of an impulsive force on the momentum of an object, and the effect of lengthening the time, stopping, or rebounding from the collision.		
	co-relate Newton's third law of motion and conservation of momentum.	G11PhU03P08	Law of Conservation of Momentum
	describe that while momentum of a system is always conserved in interaction between bodies some change in K.E. usually takes place.	G11PhU03P09	Elastic and Inelastic Collision (Derivation)
	solve different problems of elastic and inelastic collisions between two bodies in one dimension by using law of conservation of momentum.	G11PhU03P10	Examples of Elastic Collision

	identify that for a perfectly elastic collision, the relative speed of approach is equal to the relative speed of separation. describe and use of the concept of weight as the effect of a gravitational field on a mass. describe that momentum is conserved in all situations. differentiate between explosion and collision (objects move apart instead of coming nearer).	G11PhU03P11	Explosion, Momentum and Weight
Work and Energy	describe the concept of work in terms of the product of force F and displacement d in the direction of force (Work as scalar product of F and d). distinguish between positive, negative and zero work with suitable examples.	G11PhU04P01	Work
	explain gravitational field as an example of field of force and define gravitational field strength as force per unit mass at a given point.	G11PhU04P02	Work Done Against Gravitational Force
	prove that gravitational field is a conservative field.	G11PhU04P03	Conservative Field Part 1
	compute and show that the work done by gravity as a mass 'm' is moved from one given point to another does not depend on the path followed.	G11PhU04P04	Conservative Field Part 2
	describe that the gravitational PE is measured from a reference level and can be positive or negative, to denote the orientation from the reference level. define potential at a point as work done in bringing unit mass from infinity to that point. differentiate conservative and non conservative forces giving examples of each.	G11PhU04P05	Absolute Potential Energy
	express power as scalar product of force and velocity. utilize work – energy theorem in a resistive medium to solve problems.	G11PhU04P07	Power and Work Energy Principle
	discuss and make a list of limitations of some conventional sources of energy. describe the potentials of some non-conventional sources of energy.	G11PhU04P09	Non-Conventional Energy Sources
	define angular displacement, angular velocity and angular acceleration and	G11PhU05P01	

Rotational and Circular Motion	express angular displacement in radians.		Uniform Circular Motion and Centripetal Force
	solve problems by using S= r θ and v=rω .		
	describe qualitatively motion in a curved path due to a perpendicular force.		
	derive and use centripetal acceleration $a = r\omega^2$, $a = v^2 /r$.	G11PhU05P02	Centripetal Acceleration
	state and use of equations of angular motion to solve problems involving rotational motions.		
	describe situations in which the centripetal acceleration is caused by a tension force, a frictional force, a gravitational force, or a normal force.	G11PhU05P03	Centripetal Force
	solve problems using centripetal force $F = mr\omega^2$, $F = mv^2 /r$.		
	explain that satellites can be put into orbits round the earth because of the gravitational force between the earth and the satellite.	G11PhU05P05	Orbital Velocity
	define the term orbital velocity and derive relationship between orbital velocity, the gravitational constant, mass and the radius of the orbit.		
	explain that the objects in orbiting satellites appear to be weightless.	G11PhU05P06	Weightlessness and Artificial Gravity
	describe how artificial gravity is created to counter balance weightless.		
	analyze that satellites can be used to send information between places on the earth which are far apart, to monitor conditions on earth , including the weather, and to observe the universe without the atmosphere getting in the way.	G11PhU05P07	Geostationary Orbits
	describe that communication satellites are usually put into orbit high above the equator and that they orbit the earth once a day so that they appear stationary when viewed from earth.		
	define moment of inertia of a body and angular momentum.	G11PhU05P08	Moment of Inertia
	derive a relation between torque, moment of inertia and angular acceleration.	G11PhU05P09	Moment of Inertia Problemso
	use the formulae of moment of inertia of various bodies for solving problems.		

	explain conservation of angular momentum as a universal law and describe examples of conservation of angular momentum.	G11PhU05P10	Angular Momentum
Fluid Dynamics	define the terms: steady (streamline or laminar) flow, incompressible flow and non viscous flow as applied to the motion of an ideal fluid.	G11PhU06P01	Types of Flow
	explain that at a sufficiently high velocity, the flow of viscous fluid undergoes a transition from laminar to turbulence conditions.		
	describe that the majority of practical examples of fluid flow and resistance to motion in fluids involve turbulent rather than laminar conditions.		
	describe equation of continuity Av = Constant, for the flow of an ideal and incompressible fluid and solve problems using it.	G11PhU06P02	Equation of Continuity
	identify that the equation of continuity is a form of the principle of conservation of mass.		
	describe that the pressure difference can arise from different rates of flow of a fluid (Bernoulli effect).	G11PhU06P03	Bernoulli's Equation Part 1
	derive Bernoulli equation in the form P + ½ ρv2 + ρgh = constant for the case of horizontal tube of flow.	G11PhU06P04	Bernoulli's Equation Part 2
	interpret and apply Bernoulli Effect in the: filter pump, Venturi meter, in, atomizers, flow of air over an aerofoil and in blood physics.	G11PhU06P06	Venturi Meter, Viscous Drag and Stroke’s Law
	describe that viscous forces in a fluid cause a retarding force on an object moving through it.		
	explain how the magnitude of the viscous force in fluid flow depends on the shape and velocity of the object.		
	apply dimensional analysis to confirm the form of the equation F = Aηrv where ‘A’ is a dimensionless constant (Stokes’ Law) for the drag force under laminar conditions in a viscous fluid.		
	apply Stokes’ law to derive an expression for terminal velocity of spherical body falling through a viscous fluid.	G11PhU06P08	Terminal Velocity
Oscillations	describe simple examples of free oscillations.	G11PhU07P01	Oscillations and Simple Harmonic Motion
	describe necessary conditions for execution of simple harmonic motions.		

	define the terms amplitude, period, frequency, angular frequency and phase difference and express the period in terms of both frequency and angular frequency.	G11PhU07P02	Vibratory Motion
	describe that when an object moves in a circle, the motion of its projection on the diameter of the circles is SHM.	G11PhU07P03	Uniform Circular Motion and Simple Harmonic Motion
	identify and use the equation; $a= - \omega 2 x$ as the defining equation of SHM.		
	prove that the motion of mass attached to a spring is SHM.	G11PhU07P04	Mass-Spring System and Simple Harmonic Motion
	describe the interchanging between kinetic energy and potential energy during SHM.		
	analyze the motion of a simple pendulum is SHM and calculate its time period.	G11PhU07P05	Simple Pendulum
Waves	describe what is meant by wave motion as illustrated by vibrations in ropes, springs and ripple tank.	G11PhU08P01	Waves
	compare transverse and longitudinal waves.		
	define and apply the following terms to the wave model; medium, displacement, amplitude, period, compression, rarefaction, crest, trough, wavelength, velocity.	G11PhU08P02	Terms Related to Waves
	identify that sound waves are vibrations of particles in a medium.		
	solve problems using the equation: $v = f\lambda$.		
	describe that energy is transferred due to a progressive wave.	G11PhU08P03	Energy in Waves
	describe the principle of superposition of two waves from coherent sources.	G11PhU08P06	Superposition of Sound Waves
	describe the phenomenon of interference of sound waves.	G11PhU08P07	Interference of Sound and Beats
	describe the phenomenon of formation of beats due to interference of non coherent sources.		
Physical Optics	describe light waves as a part of electromagnetic waves spectrum.	G11PhU09P01	Introduction to Physical Optics
	describe the concept of wave front.		
	state Huygen's principle and use it to construct wave front after a time interval.	G11PhU09P02	

	state the necessary conditions to observe interference of light.		Huygen's Principle and Interference of Light
	describe Young's double slit experiment and the evidence it provides to support the wave theory of light.	G11PhU09P03	Young's Double Slit Experiment (Part 1)
		G11PhU09P03A	Young's Double Slit Experiment (Part 2)
		G11PhU09P03B	Young's Double Slit Experiment (Part 3)
	describe the parts and working of Michleson Interferometer and its uses.	G11PhU09P05	Michelson Interferometer
	explain diffraction and identify that interference occurs between waves that have been diffracted.	G11PhU09P06	Diffraction of Light
	describe that diffraction of light is evidence that light behaves like waves.		
	describe and explain diffraction at a narrow slit.	G11PhU09P07	Diffraction Due to a Narrow Slit
	describe the use of a diffraction grating to determine the wavelength of light and carry out calculations using $d\sin\theta=n\lambda$.	G11PhU09P08	Diffraction Grating
	describe the phenomena of diffraction of X-rays through crystals.	G11PhU09P09	Diffraction of X-Rays by Crystal
	explain polarization as a phenomenon associated with transverse waves.	G11PhU09P10	Polarization
	identify and express that polarization is produced by a Polaroid.		
	explain the effect of rotation of Polaroid on Polarization.		
	explain how plane polarized light is produced and detected.		
Thermodynamics	describe that thermal energy is transferred from a region of higher temperature to a region of lower temperature.	G11PhU10P01	Heat and Temperature
	describe that regions of equal temperatures are in thermal equilibrium .		
	describe that heat flow and work are two forms of energy transfer between systems and calculate heat being transferred.		
	describe the mechanical equivalent of heat concept, as it was historically developed, and solve problems involving work being done and temperature change.		
	explain that internal energy is determined by the state of the system and that it can be expressed as the		

	sum of the random distribution of kinetic and potential energies associated with the molecules of the system.		
	define thermodynamics and various terms associated with it.	G11PhU10P02	First Law of Thermodynamics and Isobaric Process
	relate a rise in temperature of a body to an increase in its internal energy.		
	calculate work done by a thermodynamic system during a volume change.		
	describe the first law of thermodynamics expressed in terms of the change in internal energy, the heating of the system and work done on the system.		
	explain that first law of thermodynamics expresses the conservation of energy.		
	define the terms, specific heat and molar specific heats of a gas.	G11PhU10P04	Molar Specific Heat Capacity
	apply first law of thermodynamics to derive Cp – Cv = R.		
	state the working principle of heat engine.	G11PhU10P05	Carnot Engine
	explain the working principle of Carnot's engine		
	explain that the efficiency of a Carnot engine is independent of the nature of the working substance and depends on the temperatures of hot and cold reservoirs.		
	describe the concept of reversible and irreversible processes.	G11PhU10P08	Entropy
	describe that change in entropy is positive when heat is added and negative when heat is removed from the system.		
	explain that increase in temperature increases the disorder of the system.		
	explain that increase in entropy means degradation of energy.		
	explain that energy is degraded during all natural processes.		
	identify that system tend to become less orderly over time.		
Forces and Motion	describe the equation tanθ = v2 /rg, relating banking angle θ to the speed v of the vehicle and the radius of curvature r	G11PhU03P06	Banking of Roads

work and energy	utilize work – energy theorem in a resistive medium to solve problems.	G11PhU04P09A	Work-Energy Theorem Part 1
	utilize work – energy theorem in a resistive medium to solve problems.	G11PhU04P09B	Work-Energy Theorem Part 2
Fluid Dynamics	interpret and apply Bernoulli Effect in the: filter pump, Venturi meter, in, atomizers, flow of air over an aerofoil and in blood physics.	G11PhU06P05	Bernoulli's Equation Part 3
Waves	explain that speed of sound depends on the properties of medium in which it propagates and describe Newton's formula of speed of waves.	G11PhU08P04	Speed of Sound Wave
	Identify the factors on which speed of sound in air depends.		
	Describe the Laplace correction in Newton's formula for speed of sound in air.	G11PhU08P05	Laplace Correction
	explain the formation of stationary waves using graphical method	G11PhU08P08	Stationary Waves
	define the terms, node and antinodes.		
	describe modes of vibration of strings.	G11PhU08P09	Modes of Vibration of Strings
	describe formation of stationary waves in vibrating air columns.	G11PhU08P10	Stationary Waves in Vibrating Air Columns
	describe practical examples of free and forced oscillations (resonance).	G11PhU08P11	Resonance
	describe graphically how the amplitude of a forced oscillation changes with frequency near to the natural frequency of the system.		
	describe practical examples of damped oscillations with particular reference to the efforts of the degree of damping and the importance of critical damping in cases such as a car suspension system.		

	describe qualitatively the factors which determine the frequency response and sharpness of the resonance.		
Thermodynamics	define the terms, specific heat and molar specific heats of a gas.	G11PhU10P03	Heat Capacity and Specific Heat Capacity
	state and explain second law of thermodynamics.	G11PhU10P06	Second Law of Thermodynamics
	describe that refrigerator is a heat engine operating in reverse as that of an ideal heat engine.	G11PhU10P07	Refrigerator

f. Urdu

Unit	SLO	Package Code	Package Name
اپنی مدد آپ	نئے متن کو سمجھ کر استحسانی سوالات کا جواب تحریر کر سکے اور عبارت کا عنوان تجویز کر سکے	G11UrU01P01	اپنی مدد آپ حصہ 1
	کسی ادبی یا علمی تحریر میں فنی و فکری تجزیہ کر سکے۔فنی و فکری تجزیہ کر کے اپنی رائے دے سکے اور مطلوبہ صنف میں اس کے مقام کا اصولی تعین کر سکے۔	G11UrU01P02	اپنی مدد آپ حصہ 2
	کسی ادب پارے کے مرکزی خیال کو بیان کرسکے اور تشریح کرسکے نیز اہم نکات کا خلاصہ لکھ سکے۔	G11UrU01P03	اپنی مدد آپ حصہ 3
	۔ کسی علمی مضمون میں استعمال کیے گئے الفاظ و اصطلاحات کو سمجھ سکے اور نفس مضمون کا ادراک کرسکے۔	G11UrU01P04	اپنی مدد آپ حصہ 4
	کسی ادب پارے کے حسن و قبح کا اندازہ کر سکے۔	G11UrU01P05	اپنی مدد آپ حصہ 5
جھوٹے آدمی	ادب پڑھ کر تخلیقی صلاحیتیں پیدا کرسکے۔مطبوعہ و غیر مطبوعہ مواد لکھ اور پڑھ سکے ۔ خود بھی اس قسم کی تحریر لکھ سکے۔	G11UrU02P06	جھوٹے آدمی حصہ 1
	۔بات درمیان سے سن کر سیاق و سباق کو سمجھنے اور موضوع بیان کرنے کے قابل ہو سکے۔	G11UrU02P07	جھوٹے آدمی حصہ 2
نظریۂ پاکستان	سن کر طویل کلام کے اہم نکات، جملے،مصرع یا اشعار یاد رکھ سکے۔	G11UrU03P08	نظریۂ پاکستان حصہ 1
	احساس، جذبے اور تاثر کے حوالے سے مجازی زبان سن کر حقیقت کا ادراک کر سکے۔	G11UrU03P09	نظریۂ پاکستان حصہ 2
	استحسان اور تنقیدی گفتگو سن کر سمجھ سکے اور اپنی رائے دے پائے۔	G11UrU03P10	نظریۂ پاکستان حصہ 3
پاکستانی قومیت کا مسئلہ	اخبارات اور جرائد میں خبروں ،فیچروں،اداریوں،رپورٹوں اشتہاروں اور خطوط بنا م مدیر کو ان میں پوشیدہ معنی اور معنی پر اثرانداز ہونے کے حوالے سے سمجھ کر پڑھ سکے۔	G11UrU04P11	پاکستانی قومیت کا مسئلہ حصہ 1
	۔ کسی بھی عبارت کو رفتار مطالعہ کے ساتھ پڑھ سکے۔	G11UrU04P12	پاکستانی قومیت کا مسئلہ حصہ 2
	۔ کسی بھی عبارت کو رفتار مطالعہ کے ساتھ پڑھ سکے۔	G11UrU04P13	پاکستانی قومیت کا مسئلہ حصہ 3
	کسی ادبی یا علمی تحریر میں فنی و فکری تجزیہ کر سکے۔فنی و فکری تجزیہ کر کے اپنی رائے دے سکے اور مطلوبہ صنف میں اس کے مقام کا اصولی تعین کر سکے۔	G11UrU04P14	پاکستانی قومیت کا مسئلہ حصہ 4
کچھ ادب کے بارے میں	کسی ادب پارے کے مرکزی خیال کو بیان کرسکے اور تشریح کرسکے نیز اہم نکات کا خلاصہ لکھ سکے۔	G11UrU05P15	کچھ ادب کے بارے میں حصہ 1
	۔ کسی علمی مضمون میں استعمال کیے گئے الفاظ و اصطلاحات کو سمجھ سکے اور نفس مضمون کا ادراک کرسکے۔	G11UrU05P16	کچھ ادب کے بارے میں حصہ 2
	کسی ادب پارے کے حسن و قبح کا اندازہ کر سکے۔	G11UrU05P17	کچھ ادب کے بارے میں حصہ 3

لمحہِ فکریہ حصہ 1	G11UrU06P18	۔بات درمیان سے سن کر سیاق و سباق کو سمجھنے اور موضوع بیان کرنے کے قابل ہو سکے۔	لمحہِ فکریہ
لمحہِ فکریہ حصہ 2	G11UrU06P19	سن کر طویل کلام کے اہم نکات، جملے،مصرعے یا اشعار یاد رکھ سکے۔	
لمحہِ فکریہ حصہ 3	G11UrU06P20	احساس، جذبے اور تاثر کے حوالے سے مجازی زبان سن کر حقیقت کا ادراک کر سکے۔	
داروغہ جی کی پانچوں گھی میں حصہ 1	G11UrU07P21	استحسان اور تنقیدی گفتگو سن کر سمجھ سکے اور اپنی رائے دے پائے۔	داروغہ جی کی پانچوں گھی میں
داروغہ جی کی پانچوں گھی میں حصہ 2	G11UrU07P22	بات درمیان سے سن کر بتا دینے پر قادر ہو سکے۔	
داروغہ جی کی پانچوں گھی میں حصہ 3	G11UrU07P23	طویل کلام کو اپنے حافظے سے تسلسل کے ساتھ دہرا سکے۔	
داروغہ جی کی پانچوں گھی میں حصہ 4	G11UrU07P24	مجازی، ادبی اور اصطلاحی محضر میں گفتگو کر سکے۔	
آنگن حصہ 1	G11UrU08P25	کسی تحریر پر استحسانی/تنقیدی گفتگو کر سکے۔	آنگن
آنگن حصہ 2	G11UrU08P26	اخبارات اور جرائد میں خبروں ،فیچروں،اداریوں،رپورٹوں اشتہاروں اور خطوط بنا م مدیر کو ان میں پوشیدہ معنی اور معنی پر اثرانداز ہونے کے حوالے سے سمجھ کر پڑھ سکے۔	
آنگن حصہ 3	G11UrU08P27	۔ کسی بھی عبارت کو رفتار مطالعہ کے ساتھ پڑھ سکے۔	
خوب صورت بلا حصہ 1	G11UrU09P28	۔ کسی بھی عبارت کو رفتار مطالعہ کے ساتھ پڑھ سکے۔	خوب صورت بلا
خوب صورت بلا حصہ 2	G11UrU09P29	کسی ادبی یا علمی تحریر میں فنی و فکری تجزیہ کر سکے۔فنی و فکری تجزیہ کر کے اپنی رائے دے سکے اور مطلوبہ صنف میں اس کے مقام کا اصولی تعین کر سکے۔	
تعلیم بالغاں حصہ 1	G11UrU10P30	کسی ادب پارے کے مرکزی خیال کو بیان کرسکے اور تشریح کرسکے نیز اہم نکات کا خلاصہ لکھ سکے۔	تعلیم بالغاں
تعلیم بالغاں حصہ 2	G11UrU10P31	۔ کسی علمی مضمون میں استعمال کیے گئے الفاظ و اصطلاحات کو سمجھ سکے اور نفس مضمون کا ادراک کرسکے۔	
تعلیم بالغاں حصہ 3	G11UrU10P32	کسی ادب پارے کے حسن و قبح کا اندازہ کر سکے۔	
شیراز اور کنار آب حصہ 1	G11UrU11P33	۔بات درمیان سے سن کر سیاق و سباق کو سمجھنے اور موضوع بیان کرنے کے قابل ہو سکے۔	شیراز اور کنار آب
شیراز اور کنار آب حصہ 2	G11UrU11P34	سن کر طویل کلام کے اہم نکات، جملے،مصرعے یا اشعار یاد رکھ سکے۔	
شیراز اور کنار آب حصہ 3	G11UrU11P35	احساس، جذبے اور تاثر کے حوالے سے مجازی زبان سن کر حقیقت کا ادراک کر سکے۔	
شیراز اور کنار آب حصہ 4	G11UrU11P36	استحسان اور تنقیدی گفتگو سن کر سمجھ سکے اور اپنی رائے دے پائے۔	
روم: زندہ شہر حصہ 1	G11UrU12P37	بات درمیان سے سن کر بتا دینے پر قادر ہو سکے۔	روم: زندہ شہر
روم: زندہ شہر حصہ 2	G11UrU12P38	طویل کلام کو اپنے حافظے سے تسلسل کے ساتھ دہرا سکے۔	
خطوط غالب حصہ 1	G11UrU13P39	کسی ادب پارے کے حسن و قبح کا اندازہ کر سکے۔	خطوط غالب

خطوط غالب حصہ 2	G11UrU13P40	۔بات درمیان سے سن کر سیاق و سباق کو سمجھنے اور موضوع بیان کرنے کے قابل ہو سکے۔	
خطوط غالب حصہ 3	G11UrU13P41	سن کر طویل کلام کے اہم نکات، جملے،مصرعے یا اشعار یاد رکھ سکے۔	
مکاتیب اقبال حصہ 1	G11UrU14P42	احساس، جذبے اور تاثر کے حوالے سے مجازی زبان سن کر حقیقت کا ادراک کر سکے۔	مکاتیب اقبال
مکاتیب اقبال حصہ 2	G11UrU14P43	استحسان اور تنقیدی گفتگو سن کر سمجھ سکے اور اپنی رائے دے پائے۔	
مکاتیب اقبال حصہ 3	G11UrU14P44	بات درمیان سے سن کر بتا دینے پر قادر ہو سکے۔	
لالچی وزیر	G11UrU15P45	طویل کلام کو اپنے حافظے سے تسلسل کے ساتھ دہرا سکے۔	لالچی وزیر۔۔۔
اخلاص حصہ 1	G11UrU16P46	نئے متن کو سمجھ کر استحسانی سوالات کا جواب تحریر کر سکے اور عبارت کا عنوان تجویز کر سکے	اخلاص
اخلاص حصہ 2	G11UrU16P47	کسی ادبی یا علمی تحریر میں فنی و فکری تجزیہ کر سکے۔فنی و فکری تجزیہ کر کے اپنی رائے دے سکے اور مطلوبہ صنف میں اس کے مقام کا اصولی تعین کر سکے۔	
حمد حصہ 1	G11UrU17P48	کسی ادب پارے کے مرکزی خیال کو بیان کرسکے اور تشریح کرسکے نیز اہم نکات کا خلاصہ لکھ سکے۔	حمد
حمد حصہ 2	G11UrU17P49	۔ کسی علمی مضمون میں استعمال کیے گئے الفاظ و اصطلاحات کو سمجھ سکے اور نفس مضمون کا ادراک کرسکے۔	
حمد حصہ 3	G11UrU17P50	کسی ادب پارے کے حسن و قبح کا اندازہ کر سکے۔	
نعت حصہ 1	G11UrU18P51	ادب پڑھ کر تخلیقی صلاحیتیں پیدا کرسکے۔مطبوعہ و غیر مطبوعہ مواد لکھ اور پڑھ سکے ۔ خود بھی اس قسم کی تحریر لکھ سکے۔	نعت
نعت حصہ 2	G11UrU18P52	۔بات درمیان سے سن کر سیاق و سباق کو سمجھنے اور موضوع بیان کرنے کے قابل ہو سکے۔	
نعت حصہ 3	G11UrU18P53	سن کر طویل کلام کے اہم نکات، جملے،مصرعے یا اشعار یاد رکھ سکے۔	
شہر آشوب حصہ 1	G11UrU19P54	احساس، جذبے اور تاثر کے حوالے سے مجازی زبان سن کر حقیقت کا ادراک کر سکے۔	شہر آشوب
شہر آشوب حصہ 2	G11UrU19P55	استحسان اور تنقیدی گفتگو سن کر سمجھ سکے اور اپنی رائے دے پائے۔	
شہر آشوب حصہ 3	G11UrU19P56	اخبارات اور جرائد میں خبروں ،فیچروں،اداریوں،رپورٹوں اشتہاروں اور خطوط بنا م مدیر کو ان میں پوشیدہ معنی اور معنی پر اثرانداز ہونے کے حوالے سے سمجھ کر پڑھ سکے۔	
شہزادے کا چھت پر حصہ 1	G11UrU20P57	۔ کسی بھی عبارت کو رفتار مطالعہ کے ساتھ پڑھ سکے۔	شہزادے کا چھت پر
شہزادے کا چھت پر حصہ 2	G11UrU20P58	۔ کسی بھی عبارت کو رفتار مطالعہ کے ساتھ پڑھ سکے۔	
در مراد حصہ 1	G11UrU21P59	کسی ادبی یا علمی تحریر میں فنی و فکری تجزیہ کر سکے۔فنی و فکری تجزیہ کر کے اپنی رائے دے سکے اور مطلوبہ صنف میں اس کے مقام کا اصولی تعین کر سکے۔	در مراد
در مراد حصہ 2	G11UrU21P60	کسی ادب پارے کے مرکزی خیال کو بیان کرسکے اور تشریح کرسکے نیز اہم نکات کا خلاصہ لکھ سکے۔	
در مراد حصہ 3	G11UrU21P61	۔ کسی علمی مضمون میں استعمال کیے گئے الفاظ و اصطلاحات کو سمجھ سکے اور نفس مضمون کا ادراک کرسکے۔	

در مراد حصہ 4	G11UrU21P62	کسی ادب پارے کے حسن و قبح کا اندازہ کر سکے۔	
تحت فرس پر علی اکبر حصہ 1	G11UrU22P63	۔بات درمیان سے سن کر سیاق و سباق کو سمجھنے اور موضوع بیان کرنے کے قابل ہو سکے۔	تحت فرس پر علی اکبر
تحت فرس پر علی اکبر حصہ 2	G11UrU22P64	سن کر طویل کلام کے اہم نکات، جملے،مصرعے یا اشعار یاد رکھ سکے۔	
امید حصہ 1	G11UrU23P65	احساس، جذبے اور تاثر کے حوالے سے مجازی زبان سن کر حقیقت کا ادراک کر سکے۔	امید
امید حصہ 2	G11UrU23P66	استحسان اور تنقیدی گفتگو سن کر سمجھ سکے اور اپنی رائے دے پائے۔	
نصیحت اخلاقی حصہ 1	G11UrU24P67	بات درمیان سے سن کر بتا دینے پر قادر ہو سکے۔	نصیحت اخلاقی
نصیحت اخلاقی حصہ 2	G11UrU24P68	طویل کلام کو اپنے حافظے سے تسلسل کے ساتھ دہرا سکے۔	
جلوہ سحر حصہ 1	G11UrU25P69	مجازی، ادبی اور اصطلاحی محضر میں گفتگو کر سکے۔	جلوہ سحر---
جلوہ سحر حصہ 2	G11UrU25P70	کسی تحریر پر استحسانی/تنقیدی گفتگو کر سکے۔	
جلوہ سحر حصہ 3	G11UrU25P71	اخبارات اور جرائد میں خبروں ،فیچروں،اداریوں،رپورٹوں اشتہاروں اور خطوط بنا م مدیر کو ان میں پوشیدہ معنی اور معنی پر اثرانداز ہونے کے حوالے سے سمجھ کر پڑھ سکے۔	
پرانا کوٹ حصہ 1	G11UrU26P72	۔ کسی بھی عبارت کو رفتار مطالعہ کے ساتھ پڑھ سکے۔	پرانا کوٹ
پرانا کوٹ حصہ 2	G11UrU26P73	۔ کسی بھی عبارت کو رفتار مطالعہ کے ساتھ پڑھ سکے۔	
یہ سڑکیں حصہ 1	G11UrU27P74	کسی ادبی یا علمی تحریر میں فنی و فکری تجزیہ کر سکے۔فنی و فکری تجزیہ کر کے اپنی رائے دے سکے اور مطلوبہ صنف میں اس کے مقام کا اصولی تعین کر سکے۔	یہ سڑکیں---
یہ سڑکیں حصہ 2	G11UrU27P75	کسی ادب پارے کے مرکزی خیال کو بیان کرسکے اور تشریح کرسکے نیز اہم نکات کا خلاصہ لکھ سکے۔	
قطعات حصہ 1	G11UrU28P76	۔ کسی علمی مضمون میں استعمال کیے گئے الفاظ و اصطلاحات کو سمجھ سکے اور نفس مضمون کا ادراک کرسکے۔	قطعات
قطعات حصہ 2	G11UrU28P77	کسی ادب پارے کے حسن و قبح کا اندازہ کر سکے۔	
غزلیات میرتقی میر حصہ 1	G11UrU29P78	۔بات درمیان سے سن کر سیاق و سباق کو سمجھنے اور موضوع بیان کرنے کے قابل ہو سکے۔	غزلیات میرتقی میر
غزلیات میرتقی میر حصہ 2	G11UrU29P79	سن کر طویل کلام کے اہم نکات، جملے،مصرعے یا اشعار یاد رکھ سکے۔	
غزلیات میرتقی میر حصہ 3	G11UrU29P80	احساس، جذبے اور تاثر کے حوالے سے مجازی زبان سن کر حقیقت کا ادراک کر سکے۔	
غزلیات میرتقی میر حصہ 4	G11UrU29P81	استحسان اور تنقیدی گفتگو سن کر سمجھ سکے اور اپنی رائے دے پائے۔	
غزل خواجہ میر درد حصہ 1	G11UrU30P82	بات درمیان سے سن کر بتا دینے پر قادر ہو سکے۔	غزل خواجہ میر درد
غزل خواجہ میر درد حصہ 2	G11UrU30P83	طویل کلام کو اپنے حافظے سے تسلسل کے ساتھ دہرا سکے۔	

غزل خواجہ میر درد حصہ 3	G11UrU30P84	سن کر طویل کلام کے اہم نکات، جملے،مصرعے یا اشعار یاد رکھ سکے۔	
غزل مصحفی حصہ 1	G11UrU31P85	احساس، جذبے اور تاثر کے حوالے سے مجازی زبان سن کر حقیقت کا ادراک کر سکے۔	غزل مصحفی
غزل مصحفی حصہ 2	G11UrU31P86	استحسان اور تنقیدی گفتگو سن کر سمجھ سکے اور اپنی رائے دے پائے۔	
غزل مصحفی حصہ 3	G11UrU31P87	بات درمیان سے سن کر بتا دینے پر قادر ہو سکے۔	
غزلیات غالب حصہ 1	G11UrU32P88	سن کر طویل کلام کے اہم نکات، جملے،مصرعے یا اشعار یاد رکھ سکے۔	غزلیات غالب
غزلیات غالب حصہ 2	G11UrU32P89	احساس، جذبے اور تاثر کے حوالے سے مجازی زبان سن کر حقیقت کا ادراک کر سکے۔	
غزلیات غالب حصہ 3	G11UrU32P90	استحسان اور تنقیدی گفتگو سن کر سمجھ سکے اور اپنی رائے دے پائے۔	
غزلیات غالب حصہ 4	G11UrU32P91	بات درمیان سے سن کر بتا دینے پر قادر ہو سکے۔	
غزلیات غالب حصہ 5	G11UrU32P92	۔بات درمیان سے سن کر سیاق و سباق کو سمجھنے اور موضوع بیان کرنے کے قابل ہو سکے۔	
معاون افعال	G11UrU33P93	مہارت: زبان شناسی مرکب مصادر سے معاون افعال سیکھ سکےاور عبارت سے معاون افعال کو تلاش کر سکے۔	معاون افعال
		مہارت: زبان شناسی متعلق فعل کی تعریف و استعمال سے آگاہ ہو سکے اور عبارت سے متعلق افعال تلاش کر سکے۔	متعلق فعل
متعلق فعل, قواعد	G11UrU34P94	۔لغت کو اشتقاق،مشتقات،وضعی و لغوی معنی،سلینگ، بول چال،اصطلاحی معنی اور دیگر ایسی اقسام کے حوالے سے استعمال کر سکے۔خاص علوم اور پیشوں کی اصطلاحی لغات کا استعمال سیکھ سکے 2۔ تکنیکی قواعد زبان کا ادراک کر سکے اور لسانی انتخاب لفظی کے قواعد سمجھ سکے	
تشبیہ، استعارہ	G11UrU35P95	1۔مکمل علم بیان، بدیع اور صنعتوں کی تعریف کر سکے۔ اشعار میں مثالیں تلاش کر سکے۔ 2۔ غلط فقرات کی صنائع بدائع کے لحاظ سے درستی کر سکے۔بدائع کے لحاظ سے مثالیں تلاش کر سکے۔ 3۔ کسی فن پارے کا علم بیان کی روشنی میں جائزہ لے سکے۔کسی فن پارے میں علم بیان یعنی تشبیہ، استعارہ، کنایہ ، مجاز مرسل وغیرہ کی شناخت کر سکے۔	تشبیہ، استعارہ، تلمیح---
تشبیہ، تلمیح	G11UrU35P96		
مجاز مرسل حصہ 1	G11UrU36P97	1۔مکمل علم بیان، بدیع اور صنعتوں کی تعریف کر سکے۔ اشعار میں مثالیں تلاش کر سکے۔ 2۔ غلط فقرات کی صنائع بدائع کے لحاظ سے درستی کر سکے۔بدائع کے لحاظ سے مثالیں تلاش کر سکے۔ 3۔ کسی فن پارے کا علم بیان کی روشنی میں جائزہ لے سکے۔کسی فن پارے میں علم بیان یعنی تشبیہ، استعارہ، کنایہ ، مجاز مرسل وغیرہ کی شناخت کر سکے۔	مجاز مرسل ،کنایہ
مجاز مرسل حصہ 2	G11UrU36P98		
صنائع بدائع حصہ 1	G11UrU37P99	مہارت : پڑھنا 1۔ ادبی تحریروں میں مجازی حوالے سے حسن بیان کی صنعتوں کو ملحوظ رکھ کر انھیں پڑھ سکے۔ مہارت:زبان شناسی 1۔مکمل علم بیان، بدیع اور صنعتوں کی تعریف کر سکے۔ اشعار میں مثالیں تلاش کر سکے۔ 2۔غلط فقرات کی صنائع بدائع کے لحاظ سے درستی کر سکے۔بدائع کے لحاظ سے مثالیں تلاش کر سکے۔	صنائع بدائع
صنائع بدائع حصہ 2	G11UrU37P100		
اصناف نثر حصہ 1	G11UrU38P101	مہارت : زبان شناسی 1۔اصناف سخن کی تعریف کرسکے اور پہچان کر استحسان کر سکے۔معنی کے لحاظ سے لغوی اور قرارداد اصطلاحی میں فرق کر سکے۔ 2۔ مختصر اصناف کے طرز بیان سے آگاہ ہو سکے۔بنیادی اسالیب بیان سے آگاہ ہو سکے۔	اصناف نثر
اصناف نثر حصہ 2	G11UrU38P102		

اصناف سخن حصہ 1	G11UrU39P103		اصناف سخن
اصناف سخن حصہ 2	G11UrU39P104		
اصناف نظم	G11UrU40P105		اصناف نظم پارٹ ٹو
روداد حصہ 1	G11UrU41P106	مہارت: لکھنا 1۔ کھیلوں اور معاشرتی مسائل پر اخبار کے مدیر/متعلقہ محکمہ کو رپورٹ لکھ کر بھیج سکے۔ 2۔ روزمرہ زندگی کے حوالے سے مفصل روداد،آنکھوں دیکھا حال لکھ سکے۔ 3۔ روزمرہ زندگی کے تجربات اور مشاہدات کے حوالے سے عمومی مضامین تحریر کر سکے۔ مہارت: بولنا 1۔بات کو دہراتے ہوئے اپنے علم اور تجربے کی روشنی میں کمی بیشی کے ساتھ بیان کر سکے۔ مہارت : انشاء پردازی 1۔ کسی بھی موضوع(ادبی،صحافتی، علمی)پر اپنے وسیع مطالعہ کی روشنی میں ترتیب، استدلال، عمدہ مثالوں کے ساتھ منظر نگاری، مکالمہ نگاری،کردار نگاری، کہانی،انشائیہ یا مضمون وغیرہ کی صورت میں تحریر پیش کر سکے۔ 2۔ کسی تخلیقی سطح کی تحریر (ڈائری، رپور تاژ،انشائیہ،کہانی ،افسانہ، نظم، غزل وغیرہ) کومناسب اصلاح کے ساتھ پیش کر سکے۔	روداد
روداد حصہ 2	G11UrU41P107		
مکالمہ نویسی	G11UrU42P108	مہارت: سننا سن کر بات/کہانی/مکالمے کی جزئیات کو ترتیب سے یاد رکھ سکے تاکہ بعد ازاں اسے دہرانے کے قابل ہو۔ مہارت: بولنا 1۔ سن کر بات/کہانی/مکالمے کو ترتیب سے دہرا سکے۔ مہارت : انشاء پردازی 1۔ کسی بھی موضوع(ادبی،صحافتی، علمی)پر اپنے وسیع مطالعہ کی روشنی میں ترتیب، استدلال، عمدہ مثالوں کے ساتھ منظر نگاری، مکالمہ نگاری،کردار نگاری، کہانی،انشائیہ یا مضمون وغیرہ کی صورت میں تحریر پیش کر سکے۔	مکالمہ نویسی
خط نویسی	G11UrU43P109	مہارت: پڑھنا 1۔اخبارات اور جرائد میں خبروں ،فیچروں،اداریوں،رپورٹوں اشتہاروں اور خطوط بنا م مدیر کو ان میں پوشیدہ معنی اور معنی پر اثرانداز ہونے کے حوالے سے سمجھ کر پڑھ سکے۔ 2۔ دفاتر سے آنے والے تبادلہ و تقرر کے احکام/درخواستیں /مراسلے سمجھ کر پڑھ سکے۔ مہارت: لکھنا 1۔ اخبارات اور رسائل میں سے کوئی نیوز اسٹوری، خط بنام مدیر یا کوئی مضمون لکھ سکے۔ نکاح نامہ پر کر سکے۔ 2۔ کھیلوں اور معاشرتی مسائل پر اخبار کے مدیر/متعلقہ محکمہ کو رپورٹ لکھ کر بھیج سکے۔ مہارت : مہارت حیات 1۔ روزمرہ زندگی کے مسائل کے حوالے سے غیر رسمی خطوط لکھ سکے۔داخلہ فارم، شناختی کارڈ فارم پُر کر سکے۔رسید لکھ سکے۔	خط نویسی
درخواست نویسی	G11UrU44P110	مہارت: پڑھنا 1۔ دفاتر سے آنے والے تبادلہ و تقرر کے احکام/درخواستیں /مراسلے سمجھ کر پڑھ سکے۔ مہارت: لکھنا 1۔ تبادلہ و ترقی کے لیے محکمانہ درخواست لکھ سکے۔	درخواست نویسی
تلخیص نگاری	G11UrU45P111	مہارت : لکھنا 1۔ طویل متون کی تلخیص کر سکے یا خلاصے کے طور پر سوالوں کے جواب دے سکے	تلخیص نگاری
کہانی نویسی	G11UrU46P112	مہارت: سننا" سن کر بات/کہانی/مکالمے کی جزئیات کو ترتیب سے یاد رکھ سکے تاکہ بعد ازاں اسے دہرانے کے قابل ہو۔ - مہارت: بولنا - سن کر بات/کہانی/مکالمے کو ترتیب سے دہرا سکے1	کہانی نویسی

		مہارت : انشاء پردازی ۔ کسی تخلیقی سطح کی تحریر (ڈائری، رپور تاژ،انشائیہ،کہانی1 افسانہ، نظم، غزل وغیرہ) کومناسب اصلاح کے ساتھ پیش کر، سکے۔ "	
قواعد تعارف, ردیف، قافیہ	G11UrU47P113	شاعری میں متعلقہ اصطلاحات کے تحت درج نکات کی شناخت کر سکے۔شاعری میں مطلع، مقطع، ردیف، قافیہ ،وزن، سادہ بحر کی شناخت کرسکے۔	مطلع ،مقطع، ، ردیف، قافیہ
مطلع ،مقطع	G11UrU47P114		

5. How to Use This Document in Class

This guide is designed to make lesson planning and classroom delivery easier. Teachers can use it in the following ways:

- **Lesson Planning:** Before each lesson, check the tables to see which video(s) cover the relevant SLO(s) for your class.
- **Classroom Instruction:** Play the appropriate video during class to illustrate concepts, reinforce learning, or introduce new topics.
- **Curriculum Alignment:** Ensure videos are paired with textbook units to maintain alignment with the Pakistan National Curriculum.
- **Revision and Reinforcement:** Use videos to review previous lessons or provide extra practice for students who need it.
- **Multi-Grade or Mixed-Level Classes:** Teachers can identify videos suitable for different grade levels and adjust classroom activities accordingly.
- **Assessment:** Each video package has a corresponding assessment, named with the same package code, which can be found in the accompanying Excel sheets. Teachers can use these assessments to check students' understanding immediately after showing the video or during lesson follow-ups.

The document is structured to make it easy to locate videos quickly, so you spend more time teaching and less time searching.

6. Ethical and Copyright Considerations

All digital content provided through this platform is procured by the Ministry of Federal Education and Professional Training and aligns with the Pakistan National Curriculum. When using these resources, teachers should keep the following points in mind:

- **Authorized Use Only:** Videos are for educational purposes within classrooms or official teaching activities. Do not distribute, copy, or upload content to public platforms without permission.
- **Attribution:** When using content in presentations or shared materials, credit the Ministry as the source.
- **Respect for Learners:** Ensure all content is used in a way that supports inclusive and respectful learning environments.
- **No Modification Without Approval:** Do not alter video content, subtitles, or audio tracks unless officially authorized.
- **Data Privacy:** Avoid sharing student information or recordings when using videos in digital or remote learning settings.

  Following these guidelines ensures that teachers use the digital content responsibly while maintaining legal and ethical standards.

7. Conclusion

This guide is designed to make using digital learning videos simple, efficient, and aligned with the Pakistan National Curriculum and Student Learning Outcomes. By following the mapping tables and video codes, teachers can quickly identify the right resources for each lesson, support student learning effectively, and plan classroom activities with confidence.

We encourage teachers to explore the videos, pair them with assessments, and integrate them into lesson plans to create engaging and interactive learning experiences.

For any questions, technical issues, or additional support, please contact the Federal Directorate of Education (FDE)'s content team or the designated focal points.

Empower your students, explore the digital resources, and make every lesson an engaging step toward learning success.